The Covariant Derivative

From General Relativity to Quantum Electrodynamics

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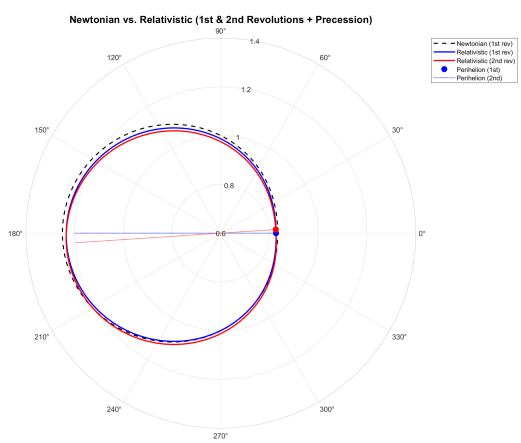
June 2025



Perihelion precession as a relativistic effect

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1 Introduction to the Project

The aim of this project is to explore the covariant derivative with a major focus on general relativity and an application to quantum electrodynamics. In order to progress further towards my goal of understanding String Theory, I have also taken some extra time to explain and go over core concepts of differential geometry in the appendices. These will also come in handy for the section on the covariant derivative. Since this project develops all the machinery required for it, I have also added a section on deriving Einstein's equations for gravity. The project flows like:

1. The Free Particle in Minkowski Space: Understanding special relativistic effects and their application to the free particle.

2. Geodesics: Developing the geodesic equation, applying it to a few special cases and gaining an understanding of how particles move in curved spacetime. This is necessary background for understanding the covariant derivative.

3. The Covariant Derivative: Developing the main part of this project and investigating its applications.

4. Deriving Einstein's Equations: A quick detour into the fundamental equations of general relativity.

5. Affine Connection in QED: Applying the covariant derivative in quantum electrodynamics.

Note that I use conformal and Weyl transformations interchangeably in this document.

2 The Free Particle in Minkowski Space

The main goal of this section is to investigate relativistic effects for the free particle and slowly introduce relativistic Lagrangian mechanics. We start by going over non-relativistic Lagrangian Mechanics. This is followed by an initial look at Lagrangian mechanics in Minkowski space. We use the special relativistic Lagrangians to solve for the free particle in Minkowski space. We also quickly look at the free particle under a constant force as an extended example of special relativistic Lagrangian Mechanics.

2.1 Non-Relativistic Lagrangian Mechanics

The Least Action Principle basically states that a classical particle follows a path that has the least action. The action is defined by:

$$S = \int_{t_1}^{t_2} dt\ L(q, \dot{q}) \tag{2.1.1}$$

Thus, to obtain the equations of motion, we must minimize the action, which equates to:

$$\delta S = 0 \tag{2.1.2}$$

To find δS, we need an appropriate expansion of the Lagrangian L, using:

$$S + \delta S = \int_{t_1}^{t_2} dt\ L(q + \delta q, \dot{q} + \dot{\delta q}) \tag{2.1.3}$$

Expanding the Lagrangian in a Taylor Series about q:

$$L(q + \delta q, \dot{q} + \dot{\delta q}) = L(q, \dot{q} + \dot{\delta q}) + \delta q \frac{\partial L(q, \dot{q} + \dot{\delta q})}{\partial q} + \cdots \tag{2.1.4}$$

Now taking this expansion and expanding it about $\dot{q}$:

$$L(q + \delta q, \dot{q} + \dot{\delta q}) = L(q, \dot{q}) + \dot{\delta q} \frac{\partial L(q, \dot{q})}{\partial \dot{q}} + \delta q \frac{\partial L(q, \dot{q})}{\partial q} + \cdots \tag{2.1.5}$$

We can substitute (2.1.5) back into (2.1.3) to give us:

$$S + \delta S = \int dt\ L(q, \dot{q}) + \int dt\ \left(\dot{\delta q} \frac{\partial L}{\partial \dot{q}} + \delta q \frac{\partial L}{\partial q} \right) + \cdots \tag{2.1.6}$$

Here, the limits have been suppressed (Its a lot of work to latex them and they aren't very important). The first term is just our original action S, so we can identify δS as:

$$\delta S = \int dt\ \left(\dot{\delta q} \frac{\partial L}{\partial \dot{q}} + \delta q \frac{\partial L}{\partial q} \right) + \cdots = \int dt\ \left(\frac{\partial L}{dq} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) + \cdots \tag{2.1.7}$$

Here, I have assumed that $\int dt\ \frac{d}{dt}(\delta q \frac{\partial L}{\partial \dot{q}})$ is 0 between our limits. The following analysis is only valid under this condition. Now, by minimizing the δS term

(imposing (2.1.2)), and neglecting the higher order terms, we get the Euler-Lagrange Equations:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0 \tag{2.1.8}$$

This is analogous to Newton's Second Law in that $\frac{\partial L}{\partial \dot{q}}$ represents the momentum and we are essentially equating the rate of change of momentum to acceleration. It is also worth noting that in classical mechanics, we generally have $L = T - U$ where T is the kinetic energy and U is the potential energy.
If we have a Hamiltonian (Which represents the total energy), we can transition between the Lagrangian and the Hamiltonian via:

$$H = \frac{\partial L}{\partial \dot{q}}\dot{q} - L \tag{2.1.9}$$

This concludes the review of Non-Relativistic Lagrangian Mechanics.

2.2 Lagrangians in Special Relativity (Minkowski Space): Part 1

Before we proceed any further it is useful to define a few terms. First is Lorentz Invariance. This condition is when the physics is the same in each inertial frame, obviously we would also require the action to be Lorentz Invariant to obey the postulates of relativity. Thus our Lagrangian must be Lorentz Invariant.
Second is the notion of proper time. This is the time measured by a particle in its rest frame. Say we have a particle traveling with a velocity v in an inertial frame spanned by x and t. Lorentz transformations are given by:

$$x' = \gamma\left(x - \frac{v}{c}ct\right); \qquad ct' = \gamma\left(ct - \frac{v}{c}x\right) \tag{2.2.1}$$

Where $\gamma = \sqrt{\frac{1}{1-\frac{v^2}{c^2}}}$ Using these equations, transforming from our inertial frame to the rest frame gives us the proper time as:

$$d\tau = \frac{dt}{\gamma} \tag{2.2.2}$$

We want to transition to the particle's rest frame to make obtaining the Lagrangian easier. The momentum of the particle is 0 in the rest frame, this gives us a rough rest-frame lagrangian:

$$L_{rest} = -H = -mc^2 \tag{2.2.3}$$

Thus, the action is given by:

$$S = -mc^2\int d\tau = -mc^2\int dt\,\sqrt{1-\frac{v^2}{c^2}} \tag{2.2.4}$$

By analogy to the non-relativistic definition of the action (2.1.1), we can conclude that our Lagrangian (In any reference frame) is given by:

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} \tag{2.2.5}$$

Since we solved this for a particle traveling at a constant velocity, we should expect this Lagrangian to be analogous to the classical Lagrangian for purely kinetic energy. To verify this, we can expand the Lagrangian in a Binomial series to impose the non-relativistic limit (($\frac{v}{c}$) is small).

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} = -mc^2\ \left(1-\frac{1}{2}\left(\frac{v}{c}\right)^2+\frac{1}{8}\left(\frac{v}{c}\right)^4+\cdots\right) = \frac{1}{2}mv^2+C \tag{2.2.6}$$

The constant C is irrelevant as it gets integrated out in the Euler-Lagrange Equation (2.1.8). Thus the Lagrangian we have found is analogous to the purely kinetic case. By analogy to the non-relativistic case, we can now say that any general relativistic Lagrangian will take the form:

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} - U \tag{2.2.7}$$

Where U is the potential energy.

2.3 Covariant Equation of Motion for the Free Particle

To get the Equation of Motion for the free particle, we simply set $U = 0$ in (2.2.7) and solve the Euler-Lagrange Equations (which can easily be verified to work for the relativistic case as well). This yields the solution:

$$\frac{d}{dt}\left(\frac{m\dot{x}}{\sqrt{1-\frac{\dot{x}^2}{c^2}}}\right) = \frac{d}{dt}(m\gamma\dot{x}) = 0 \tag{2.3.1}$$

In contrast to the classical Equation of Motion:

$$m\frac{d^2x}{dt^2} = \frac{d}{dt}(m\dot{x}) = 0 \tag{2.3.2}$$

It can be seen that the relativistic equation (2.3.1) is "corrective" in nature, we are essentially just accounting for relativistic effects like time-dilation and length-contraction through the Lorentz transform term γ which is constant in the case of the free particle. Figure 1 is a great visualization of relativistic effects at play for the free-particle. We can see that the classical particle needs a greater proper time to reach the same point that the relativistic particle doesn't. Keeping in mind that proper time is the time felt by the particle as defined in (2.2.2), we can understand how this is time dilation at work.

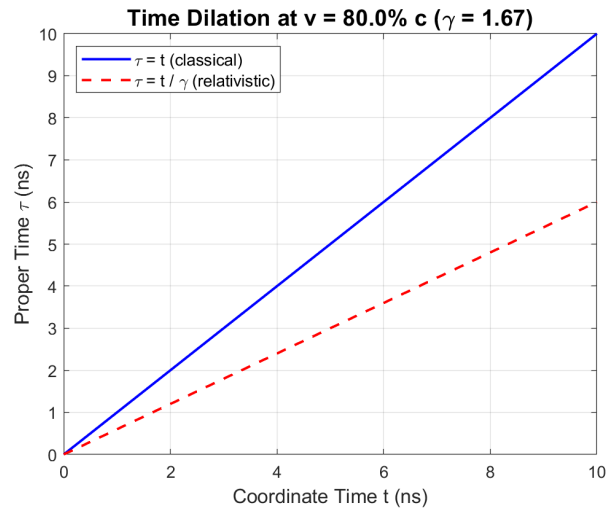

Figure 1: Time Dilation Effects in the Free-Particle

2.4 Equation of Motion for a Particle Under Constant Force

Let the force applied by α, this should be a retarding force. This means $U = \alpha x$, and our Lagrangian is given by:

$$L = -mc^2\sqrt{1 - \frac{v^2}{c^2}} - \alpha x \tag{2.4.1}$$

Solving this gives us the equation of motion,

$$m\frac{d^2x}{dt^2}\frac{1}{(1 - \frac{\dot{x}^2}{c^2})^{\frac{3}{2}}} = -\alpha \tag{2.4.2}$$

This is very evidently just $F = ma$, so we can say that the acceleration for this case will be:

$$a = -\frac{\alpha}{m}\left(1 - \frac{\dot{x}^2}{c^2}\right)^{\frac{3}{2}} \tag{2.4.3}$$

It is worth noting that the acceleration depends on the velocity, which is not constant. This is in contrast to the acceleration for the non-relativistic case:

$$a_{nr} = -\frac{\alpha}{m} \tag{2.4.4}$$

As in the previous case, this new acceleration (2.4.3) accounts for the relativistic effects such as time-dilation and length contraction. The general solution for this, starting from an initial point x_0 with velocity v_0 is:

$$x(t) = x_0 + \frac{mc^2}{\alpha}\left(\frac{1}{\sqrt{1 - \frac{v_0^2}{c^2}}} - \sqrt{\frac{1}{1 - \frac{v_0^2}{c^2}} + \frac{\alpha^2 t^2}{m^2c^2} - \frac{2v_0\alpha t}{mc^2\sqrt{1 - \frac{v_0^2}{c^2}}}}\right) \tag{2.4.5}$$

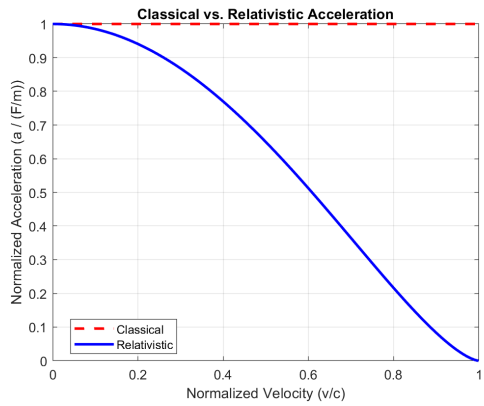

Figure 2: A Comparison of Classical and Relativistic Acceleration

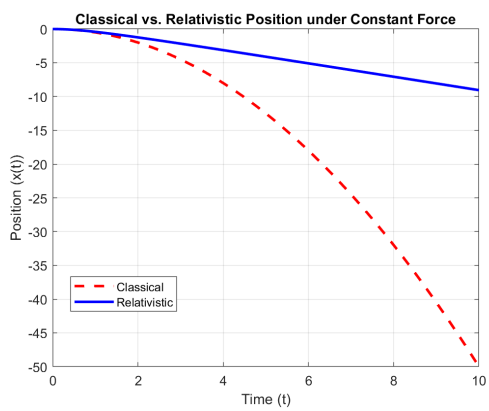

Figure 3: A Comparison of Classical and Relativistic Position

If we set $x_0 = v_0 = 0$, the equation simplifies down to:

$$x(t) = \frac{mc^2}{\alpha^2}\left(1 - \sqrt{1 + \left(\frac{\alpha t}{mc}\right)^2}\right) \tag{2.4.6}$$

Now, $\frac{mc^2}{\alpha^2}$ can be considered a constant scaling factor, ignoring this and rearranging a few terms gives us an equation of the form $(x(t) - 1)^2 - t^2 = -1$ which is can be interpreted as a hyperbolic equation. Thus the particle follows a hyperbolic trajectory for $x_0 = v_0 = 0$. The velocity for these initial conditions will be given by:

$$v(t) = \frac{\alpha ct}{\sqrt{m^2c^2 + \alpha^2 t^2}} \tag{2.4.7}$$

Figure 2 shows how the relativistic effects affect acceleration and Figure 3 plots

$x(t)$ from (2.4.5) for the relativistic and classical case.
An observation from this graph is that when travelling at the speed of light, it is impossible to feel any acceleration from a constant force. This agrees with the postulate of relativity that states that you can't go faster than light. The "response" of the relativistic particle to an external force is very dull compared to the classical particle. The figure illustrates an extreme case of this where $\frac{\alpha}{mc} = 1$.
With all of this information as a foundation, we can now move onto discussing geodesics.

3 Geodesics

Armed with an understanding of what happens to a free particle in Minkowski space, we can now look at geodesics in both flat and curved spacetime. This section draws heavy inspiration from David Tong's lecture notes. We will start by deriving the geodesic equation in the non-relativistic case, after which we move on to consider Lagrangians in flat spacetime again and finally derive the geodesic equation in curved spacetime, after which we consider a few examples.

3.1 Non-Relativistic Geodesics

The metric $g_{ij}(x)$ is used to measure the interval between two points. The general form of the interval is: $ds^2 = g_{ij}(x)x^i x^j$, for the non-relativistic case the metric is simply the identity matrix I. If we evaluate keeping $g_{ii} = 1$ in mind:

$$g_{ij}(x)x^i x^j = x_1^2 + x_2^2 + x_3^2 = x_j x^j \tag{3.1.1}$$

We want to proceed with this form instead of the equally valid $(x_1^2+x_2^2+x_3^2)$, to easily transition to curved spacetime, where the nature of spacetime is defined by the metric. The task is now to solve the Euler-Lagrange Equations (2.1.8) for a Lagrangian of the form:

$$L = \frac{m}{2} g_{ij}(x)\dot{x}^i \dot{x}^j \tag{3.1.2}$$

The first derivative will be:

$$\frac{\partial L}{\partial x^k} = \frac{m}{2}\frac{\partial}{\partial x^k}(g_{ij}(x))\dot{x}^i \dot{x}^j \tag{3.1.3}$$

The second derivative will be:

$$\frac{\partial L}{\partial \dot{x}^k} = \frac{m}{2} g_{ij}(x)\left(\dot{x}^i \frac{\partial \dot{x}^j}{\partial \dot{x}^k} + (i \leftrightarrow j)\right) = \frac{m}{2}(g_{ik}\dot{x}^i + g_{kj}\dot{x}^j) \tag{3.1.4}$$

Now, since the summation convention implies that all of these are summed over various components, doing this summation implies that the indices i and j end up being equivalent. Thus we can write:

$$\frac{\partial L}{\partial \dot{x}^k} = m g_{ki}\dot{x}^i \tag{3.1.5}$$

Taking the total time-derivative gives us:

$$\frac{d}{dt}(m g_{ik}\dot{x}^i) = m g_{ik}\frac{d^2 x^i}{dt^2} + m\frac{\partial g_{ik}}{\partial x^j}\dot{x}^i \dot{x}^j \tag{3.1.6}$$

Thus the Euler-Lagrange Equations become:

$$g_{ik}\frac{d^2 x^i}{dt^2} + \left(\frac{\partial g_{ik}}{\partial x^j} - \frac{1}{2}\frac{\partial g_{ij}}{\partial x^k}\right)\dot{x}^i \dot{x}^j = g_{ik}\frac{d^2 x^i}{dt^2} + \frac{1}{2}\left(\frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{jk}}{\partial x^i}\right)\dot{x}^i \dot{x}^j = 0 \tag{3.1.7}$$

Here, we have simply used symmetric to expand the metric. To make more sense of this step, we should expand the expression into individual indices, this shows the equivalent nature. We can now multiply the inverse metric g^{ij} and rename some indices to give us the non-relativistic geodesic equation:

$$\frac{d^2x^i}{dt^2} + \Gamma^i_{jk}\frac{\partial x^j}{\partial t}\frac{\partial x^k}{\partial t} = 0 \tag{3.1.8}$$

Where Γ^i_{jk} denotes the Christoffel symbol defined by:

$$\Gamma^i_{jk}(x) = \frac{1}{2}g^{il}\left(\frac{\partial g_{lj}}{\partial x^k} + \frac{\partial g_{lk}}{\partial x^j} - \frac{\partial g_{jk}}{\partial x^l}\right) \tag{3.1.9}$$

Note: I have re-arranged a few indices to get the form of Equation (3.1.8), this is in order to keep consistency with David Tong's Lecture Notes. The solutions to Equation (3.1.8) are called geodesics. Geodesics can be thought of as the curves representing the shortest distance between any two points.
The next step is to make this equation relativistic. However, before we can make this equation relativistic, we need to understand relativistic Lagrangian mechanics better.

3.2 Lagrangians in Minkowski Space: Part 2

Previously, we discussed that we can write the action in Minkowski space as:

$$S = -mc\int d\tau \tag{3.2.1}$$

Where, τ is the proper time as defined in (2.2.2). We know that the proper time is the actual time "felt" by the particle. Now, if a particle travels an infinitesimal distance in Minkowski space, we can say that this distance is equal to an infinitesimal change in proper time. Thus, we can also equivalently write the action as:

$$S = -mc\int\sqrt{-ds^2} = -mc\int\sqrt{-\eta_{\mu\nu}dx^\mu dx^\nu} \tag{3.2.2}$$

Where, the root and minus act as a way to only consider time-like particles.
Now, if the particle is traveling along a curve parameterized by σ, we can write the action as:

$$S = -mc\int_{\sigma_1}^{\sigma_2} d\sigma\sqrt{-\eta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}} \tag{3.2.3}$$

Writing the relativistic action this way allows us to easily verify Lorentz invariance and also reparameterization invariance. *The latter also explains why a particle is always moving through Minkowski space. Essentially, solving the action gives us 4 different equations of motion, however we have no information about the parameter σ, so we need to use two of those equations to eliminate sigma. This then means that not all coordinates x^μ are independent. Which implies that the momenta are also dependent on each other, and thus the particle*

is always moving.
If we want to make a Lagrangian that accounts for curvature, we simply make the substitution $\eta_{\mu\nu} \mapsto g_{\mu\nu}(x)$. The metric $g_{\mu\nu}(x)$ encodes all the information about the curvature of spacetime. Thus, we can write the Lagrangian for curved spacetime as:

$$S = -mc \int d\sigma \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} \tag{3.2.4}$$

For future purposes, we can define the proper time for a particle travelling along a curve as:

$$c\tau(\sigma) = \int_0^\sigma d\sigma \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} \tag{3.2.5}$$

This is all the information we need to finally get a relativistic geodesic equation.

3.3 Geodesics in Curved Spacetime

To get the relativistic geodesic equation, we must solve the Euler-Lagrange Equations for the action given in (3.2.4). Thus,

$$L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} \tag{3.3.1}$$

The process of solving this equation will be similar to that done in Section 2.1. However, the presence of the square root will make things a bit trickier. Let's start with the first derivative:

$$\frac{\partial L}{\partial x^\rho} = \frac{\partial}{\partial x^\rho}(-g_{\mu\nu}(x)\dot{x}^\mu \dot{x}^\nu) = \frac{-1}{2L} \frac{\partial g_{\mu\nu}(x)}{\partial x^\rho} \dot{x}^\mu \dot{x}^\nu \tag{3.3.2}$$

For the second derivative:

$$\frac{\partial L}{\partial \dot{x}^\rho} = \frac{\partial}{\partial \dot{x}^\rho}(-g_{\mu\nu}(x)\dot{x}^\mu \dot{x}^\nu) = \frac{-1}{2L} g_{\mu\nu}(\delta^\mu_\rho \dot{x}^\nu + (\nu \leftrightarrow \mu)) = \frac{-1}{L} g_{\mu\rho}(x)\dot{x}^\mu \tag{3.3.3}$$

Taking the total derivative gives us:

$$\frac{d}{d\sigma}\left(-\frac{1}{L} g_{\mu\rho}(x)\dot{x}^\mu\right) = -\frac{1}{L}\frac{dg_{\mu\rho}}{dx}\dot{x}^\nu \dot{x}^\mu - \frac{1}{L} g_{\mu\rho}\frac{d^2x^\mu}{d\sigma^2} + \frac{1}{L^2}\frac{dL}{d\sigma} g_{\mu\rho}\dot{x}^\mu \tag{3.3.4}$$

Putting it all together gives us the whole equation:

$$\frac{-1}{2L}\frac{\partial g_{\mu\nu}}{\partial x^\rho}\dot{x}^\mu \dot{x}^\nu + \frac{1}{L}\frac{\partial g_{\mu\rho}}{\partial x^\nu}\dot{x}^\mu \dot{x}^\nu + \frac{1}{L} g_{\mu\rho}\frac{d^2x^\mu}{d\sigma^2} = \frac{1}{L^2}\frac{dL}{d\sigma} g_{\mu\rho}(x)\dot{x}^\mu \tag{3.3.5}$$

Using the same simplification steps as the analogous non-relativistic case:

$$g_{\mu\rho}\frac{d^2x^\mu}{d\sigma^2} + \left(\frac{\partial g_{\mu\rho}}{\partial x^\nu} + \frac{\partial g_{\nu\rho}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\rho}\right)\dot{x}^\mu \dot{x}^\nu = \frac{1}{L}\frac{dL}{d\sigma} g_{\mu\rho}\dot{x}^\mu \tag{3.3.6}$$

We want to eliminate the term on the LHS now. This is where we can leverage the reparameterization invariance of the action. Essentially, we should be able to pick a parameter such that the LHS term zeros out. This parameter is the proper time, defined in Equation (3.2.5). It is easy to see that $\frac{dL}{d\tau} = 0$. Using proper time and renaming some indices gives us the geodesic equation for curved spacetime:

$$\frac{d^2x^\mu}{d\bar{\tau}^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\bar{\tau}}\frac{dx^\rho}{d\bar{\tau}} = 0 \tag{3.3.7}$$

Here, we have defined $\bar{\tau} = a\tau + b$. This is because the LHS term zeros out for any linear function of τ. These parameters a, b are called the affine parameters of the worldline. It is worth noting that we can get the same geodesic equation using the "useful" action (Derivation is essentially the same as for Equation (2.2.4)):

$$S_{useful} = \int d\tau \; g_{\mu\nu}(x)\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \tag{3.3.8}$$

Provided we enforce the time-like condition:

$$g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = -c^2 \tag{3.3.9}$$

3.4 Geodesics in R^3 and S^2

3.4.1 Geodesics in R^3

In flat space (R^3), using spherical polar coordinates gives us:

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 sin^2\theta \; d\phi^2$$

Here, the metric will be given by:

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 sin^2\theta \end{pmatrix}$$

The inverse metric will be given by:

$$g^{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & (r^2 sin^2\theta)^{-1} \end{pmatrix}$$

We can compute the Christoffel symbols. Let's start by computing $\Gamma^r_{\theta\theta}$:

$$\Gamma^r_{\theta\theta} = \frac{1}{2}g^{rr}\left(\frac{\partial g_{r\theta}}{\partial\theta} + \frac{\partial g_{r\theta}}{\partial\theta} - \frac{\partial g_{\theta\theta}}{\partial r}\right) = \frac{1}{2}(-\frac{\partial r^2}{\partial r}) = -r$$

We can then look at: $\Gamma^r_{\phi\phi}$:

$$\Gamma^r_{\phi\phi} = \frac{1}{2}g^{rr}\left(\frac{\partial g_{r\phi}}{\partial\phi} + \frac{\partial g_{r\phi}}{\partial\theta} - \frac{\partial g_{\phi\phi}}{\partial r}\right) = -\frac{1}{2}\frac{\partial}{\partial r}(r^2 sin^2\theta) = -r sin^2\theta$$

The main conclusion from these components is that having non-zero Christoffel symbol components does not imply that spacetime is curved.
We can also see that it is very annoying to compute Christoffel symbols and it is generally quicker to just solve directly using the action principle. This is what we will do in the following examples.

3.4.2 Geodesics in S^2

If we consider the two-dimensional sphere S^2. Space is truly curved here, so we shouldn't run into the same problem as the last example. The metric for the example is:

$$g_{ij} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 sin^2\theta \end{pmatrix}$$

We can get this metric by setting $r = R$ in our usual spherical coordinates. The action will be given by:

$$S = \frac{mR^2}{2} \int dt(\dot{\theta}^2 + sin^2\theta \ \dot{\phi}^2)$$

It is one of the most basic problems of the calculus of variation to solve this action. Solving the problem tells us that great circles are geodesics on the sphere. However, let's quickly check the Christoffel symbols:

$$\Gamma^{\theta}_{\theta\theta} = \frac{1}{2} g_{\theta\theta} \left(\frac{\partial g_{\theta\theta}}{\partial \theta} + \frac{\partial g_{\theta\theta}}{\partial \theta} - \frac{\partial g_{\theta\theta}}{\partial \theta} \right) = \frac{1}{2} R^2(0) = 0$$

$$\Gamma^{\theta}_{\theta\phi} = \frac{1}{2} g^{\theta\theta} \left(\frac{\partial g_{\theta\theta}}{\partial \phi} + \frac{\partial g_{\theta\phi}}{\partial \theta} - \frac{\partial g_{\theta\phi}}{\partial \theta} \right) = 0$$

$$\Gamma^{\phi}_{\phi\phi} = \frac{1}{2} g^{\phi\phi} \left(\frac{\partial g_{\phi\phi}}{\partial \phi} + \frac{\partial g_{\phi\phi}}{\partial \phi} - \frac{\partial g_{\phi\phi}}{\partial \phi} \right) = 0$$

$$\Gamma^{\phi}_{\theta\phi} = \frac{1}{2} g^{\phi\phi} \left(\frac{\partial g_{\phi\theta}}{\partial \phi} + \frac{\partial g_{\phi\phi}}{\partial \theta} - \frac{\partial g_{\theta\phi}}{\partial \phi} \right) = \frac{1}{2} R^2 sin^2\theta \left(\frac{\partial R^2 sin^2\theta}{\partial \theta} \right) = \frac{R^4}{2} (sin2\theta \ sin^2\theta)$$

We can see that S^2 has non-zero Christoffel symbols and is a genuinely curved unlike the previous example.

3.5 The Schwarzschild Metric: Geodesics

We now try to apply the Geodesic Equation (3.3.7) to investigate the motion of planets around a star. To model this, we consider planets to be particles in the spacetime curved by a star. This curvature is given by the Schwarzschild Metric:

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2(d\theta^2 + sin^2\theta \ d\phi^2) \quad (3.5.1)$$

It is worth noting that the g_{rr} component diverges at $r = R_s$, where R_s is the Schwarzschild radius:

$$R_s = \frac{2GM}{c^2} \tag{3.5.2}$$

In our case, we have $R_{star} >> R_s$. If $R_{star} \approx R_s$, we would have to deal with more physics, which is what happens with black holes.

We write the action for this metric in the form given by Equation (3.3.8). This yields:

$$S = \int d\tau \; [-A(r)c^2\dot{t}^2 + A^{-1}(r)\dot{r}^2 + r^2(\dot{\theta}^2 + sin^2\theta \; \dot{\phi}^2)] \tag{3.5.3}$$

Where $A(r) = 1 - \frac{R_s}{r}$ Here, $\dot{x} = \frac{\partial x}{\partial \tau}$. Just by looking at this action, it is visible that the ϕ and t components of the Euler-Lagrange equation will be conservation laws. Before delving into conservation laws, it's a good idea to simplify the case a little bit. This simplification is done by looking at the θ component of the Euler-Lagrange Equation:

$$\frac{d}{d\tau}(r^2\dot{\theta}) = r^2 sin\theta cos\theta \; \dot{\phi}^2 \tag{3.5.4}$$

We can now localize to a plane by setting $\theta = \frac{\pi}{2}$ and $\dot{\theta} = 0$. With this restriction in place, we can look at the ϕ component of the equations:

$$\frac{dL}{d\dot{\phi}} = 2r^2\dot{\phi} = 2l \tag{3.5.5}$$

This can be identified as conservation of angular momentum, and the 2 factor is because of how we define the action without the usual 1/2 factor. As for the time t component, we have:

$$-2E = \frac{dL}{d\dot{t}} = -2A(r)c^2\dot{t} \tag{3.5.6}$$

To see the interpretation, if we move to Minkowski space ($r \to \infty$), we will have $A(r) \approx 1$ and $\frac{dt}{d\tau} = \gamma$. This makes $E \to \gamma c^2$ as $r \to \infty$. Thus we can interpret E as rest energy per unit mass, making (3.5.6) an energy conservation law. So we have boiled the problem to a particle confined in a plane $\theta = \frac{\pi}{2}$ with both energy and angular momentum being conserved. However, before proceeding further, we should ensure that our particle is timelike. This requires the application of Equation (3.3.9). For our confined problem, we have:

$$-A(r)c^2\dot{t}^2 + A^{-1}(r)\dot{r}^2 + r^2\dot{\phi}^2 = -c^2 \tag{3.5.7}$$

With substitutions, this can be written as:

$$\frac{1}{2}\dot{r}^2 + V_{eff}(r) = \frac{1}{2}\frac{E^2}{c^2} \tag{3.5.8}$$

Where the effective potential is given by:

$$V_{eff}(stosr) = \frac{1}{2}\left(c^2 + \frac{l^2}{r^2}\right)\left(1 - \frac{2GM}{rc^2}\right) = \frac{c^2}{2} - \frac{GM}{r} + \frac{l^2}{2r^2} - \frac{l^2GM}{r^3c^2} \tag{3.5.9}$$

3.5.1 Newtonian Solution

Expanding the relativistic energy in powers of $1/c^2$ and taking the non-relativistic limit gives us the Newtonian potential:

$$\frac{1}{2}\dot{r}^2 + V_N(r) = E_N \Rightarrow V_N(r) = -\frac{GM}{r} + \frac{l^2}{2r^2} \tag{3.5.10}$$

We can identify the attractive gravitational force which scales as $1/r$ and the angular momentum barrier which scales as $1/r^2$. If $E_N < 0$, the particle is unable to escape the potential well. If $E_N \geq 0$, we have a fly-by case. This is the case where the particle has too much energy to get stuck in a planet's orbit. Here the planet comes close, almost touches the star but then goes away. The solution to this potential is given by:

$$r(\phi) = \frac{l^2}{GM}\frac{1}{1 + e\cos\phi} \Rightarrow e = \sqrt{1 + \frac{2E_N l^2}{G^2M^2}} \tag{3.5.11}$$

This is just a conic section dependent on eccentricity. It depends on E_N. Substituting $E_N \geq 0$ gives us a hyperbola with $e > 1$ and the other case gives us an ellipse with $e < 1$. The point at $\phi = 0$ is known as a perihelion.

3.5.2 Relativistic Solution

Here, we use the full potential from (3.5.9). Just by analyzing the potential, we can already identify a few relativistic corrections:

$$V'_{eff}(r) = \frac{GM}{r^2} - \frac{l^2}{r^3} + \frac{3GMl^2}{r^4c^2} = 0 \tag{3.5.12}$$

The solution of this quadratic equation depends on the value of l. There are three cases that can arise:

$$l^2 > \frac{12G^2M^2}{c^2}; l^2 < \frac{12G^2M^2}{c^2}; l^2 = \frac{12G^2M^2}{c^2} \tag{3.5.13}$$

For the first case, solving Equation (3.5.12) yields 2 critical points. These are denoted as r_+, r_- with $r_+ > r_-$. The outermost solution will be a minimum of the potential and represent the most stable orbit (potential minima) while the innermost solution is the most unstable orbit (potential maxima). This case corresponds to a high angular momentum barrier. The second case has no critical points. The third case has a saddle point at:

$$r_{ISCO} = \frac{6GM}{c^2} \tag{3.5.14}$$

This is known as the innermost stable circular orbit. No circular orbits can extend at $r < r_{ISCO}$. This does not eliminate the possibility that orbits can't extend into regions below r_{ISCO}.
The full relativistic potential can only be solved perturbatively. The first correction is given by:

$$u_1 = \frac{GM}{l^2}\left[(1 + \frac{e^2}{2}) + e\phi\sin\phi - \frac{e^2}{6}\cos 2\phi\right] \tag{3.5.15}$$

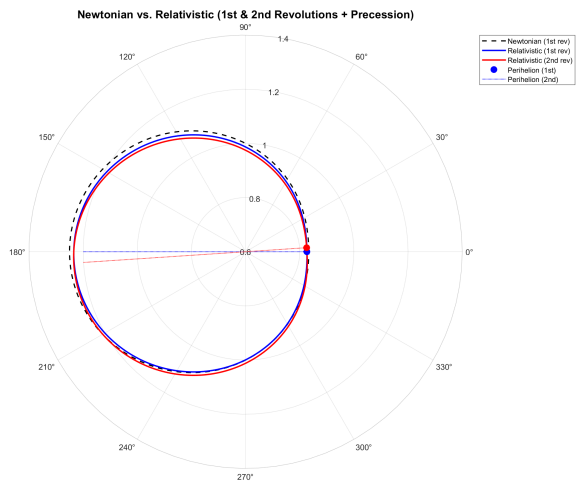

Figure 4: Perihelion Precession as a consequence of General Relativity

Here the e is identical to the previous Newtonian solution (3.5.11) and $u = 1/r$. The $e\phi \sin\phi$ term is what causes the biggest disturbance from the non-relativistic solution. Due to the perturbative nature of our solution, the perihelion of the orbit shifts by an amount δ for each "closure" of the orbit. It is important to note that the orbit doesn't strictly close because there is a different radial value for $\phi = 0$ and $\phi = 2\pi$. With this solution in hand, we can compare the non-relativistic case to the relativistic case in more detail.

3.5.3 Impact of Relativistic Corrections

The biggest difference between the two theories is perihelion precession. This is seen in Figure 4. Perihelion precession is mathematically described in the previous section with (3.5.15). We can also see from the figure that the first relativistic orbit is considerably different from the non-relativistic case. This relativistic orbit can be described (for lack of a better word) as "tighter" than its counterpart. The perihelion precession of planets can be measured experimentally and has been used as evidence to validate the predictions of general relativity.

The next major consequence of relativity is seen by looking at (3.5.9). The full potential tells us that the angular momentum barrier is finite. Thus for the case of the first case in Equation (3.5.13), it is possible for a particle to charge at a star with insanely high energies and fall off towards $r = 0$. This is impossible in the non-relativistic case. We can see this illustrated in Figure 5.

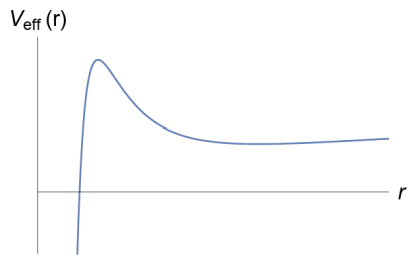

Figure 5: Potential for High Angular Momentum Case

3.5.4 A Quick Summary

The motion of planets around stars is governed by the Schwarzschild metric (3.5.1). We can calculate the geodesics for this metric either by directly solving the Euler-Lagrange equations or by using the geodesic equation (3.3.7). One of the main physical results we obtain by solving for these geodesics is perihelion precession. The perihelion is the point on the orbit where the planet is closest to the star. This point shifts by a small angle in ϕ each orbit. The prediction of perihelion precession is something that the non-relativistic case failed to achieve. Another conclusion from this analysis is that the angular momentum barrier for the relativistic case is finite. This means it is possible for a particle to charge at a star and get sucked into $r = 0$. This is in contrast to the non-relativistic case which predicts an infinite angular momentum barrier.

3.6 The Metric in De Sitter Space (A Quick Look)

De Sitter space is spacetime under the influence of a positive cosmological constant. The metric for de Sitter space is given by:

$$ds^2 = -\left(1 - \frac{r^2}{\alpha^2}\right) c^2 dt^2 + \left(1 - \frac{r^2}{\alpha^2}\right)^{-1} dr^2 + r^2(d\theta^2 + sin^2\theta \; d\phi^2) \tag{3.6.1}$$

Here, $\alpha = \sqrt{\frac{3}{\Lambda}}$ is a parameter that encodes the cosmological constant.

3.6.1 Getting the Effective Potential

To simplify this metric, we introduce the parameter $B(r) = \left(1 - \frac{r^2}{\alpha^2}\right)$. This metric simplifies to:

$$ds^2 = -B(r)c^2 dt^2 + B(r)^{-1} dr^2 + r^2(d\theta^2 + sin^2\theta \; d\phi^2) \tag{3.6.2}$$

We can write the action as:

$$S = \int d\tau \; [-B(r)c^2 \dot{t}^2 + B(r)^{-1}\dot{r}^2 + r^2(\dot{\theta}^2 + sin^2\theta \; \dot{\phi}^2)] \tag{3.6.3}$$

We can borrow the method from our previous analysis to localize our particle to a plane. By comparing this action to the one for the Schwarzschild metric

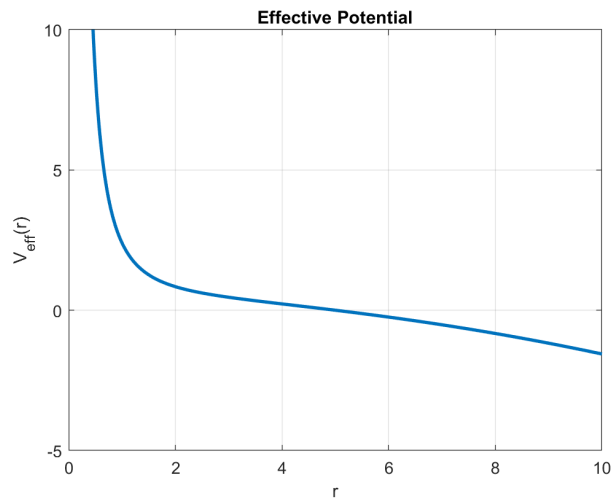

Figure 6: The Effective Potential for de Sitter Space

(3.5.3), we understand that we can localize the particle to a plane by setting $\theta = \frac{\pi}{2}$ and $\dot{\theta} = 0$.
By looking at the action, we can actually see that the t and ϕ components will be conserved. This gives us the two conservation laws:

$$\frac{dL}{d\dot{\phi}} = 2r^2\dot{\phi} = 2l \tag{3.6.4}$$

Similar to the Schwarzschild metric, we can identify this as the angular momentum from (3.5.5). Thus we can say that angular momentum is conserved. The problem here will be the time component:

$$-2\mathcal{H} = \frac{dL}{d\dot{t}} = -2B(r)c^2\dot{t} \tag{3.6.5}$$

We can't identify this as the energy. This aligns well with what de Sitter means. Equation (3.6.5) gives mathematical substance to the fact that the energy of the universe is no longer conserved in de Sitter space.

We can try solving for the general solution now. Imposing the required timelike condition gives us:

$$-B(r)c^2\dot{t}^2 + B^{-1}(r)\dot{r}^2 + r^2\dot{\phi}^2 = -c^2 \tag{3.6.6}$$

This can be simplified to:

$$\frac{1}{2}\dot{r}^2 + V_{eff}(r) = \frac{1}{2}\frac{\mathcal{H}}{c^2} \tag{3.6.7}$$

Here, the effective potential is given by:

$$V_{eff}(r) = \frac{1}{2}\left(c^2 + \frac{l^2}{r^2}\right)\left(1 - \frac{r^2}{\alpha^2}\right) = \frac{1}{2}\left(c^2 + l^2\left[\frac{1}{r^2} - \frac{1}{\alpha^2}\right] - \frac{c^2r^2}{\alpha^2}\right) \tag{3.6.8}$$

3.6.2 Investigating the Effective Potential

We can see the potential in Figure 6. It turns out that, as we go further along r, the potential continues to decrease. There are no maxima/minima for this potential (3.6.8). After reaching a certain value of r, the particle experiences an extreme angular momentum barrier. The attractive potential before the barrier is due to the curvature of de Sitter space.

Increasing the value of α increases the attractive potential faced by the particle. Using this fact, we can understand how the cosmological constant Λ influences the curvature of spacetime. Increasing Λ makes the universe 'curve upward' more so that the particle 'falls downward'. Increasing the value of α also changes the rate at which V_{eff} decreases with r. A higher value of α means a slow rate of decrease while a lower value means a greater rate of decrease.

We can also slightly understand how the geodesics in de Sitter space will differ from the ones in Minkowski space. The presence of V_{eff} alone shows the effect of curvature on a geodesic. Geodesics are a lot more interesting with the presence of curvature and depending on the choice of metric, we can even get orbits.

3.7 Summary of Geodesics

This section went over how geodesics are the paths taken by a free particle in space/spacetime. We derived the relativistic geodesic equation (3.3.7) using a few properties of the relativistic action. We then used this equation to investigate a few metrics and a few cases. An important takeaway from the case in R^3 was that a non-zero Christoffel symbol component does not necessarily imply curved spacetime.

The geodesic equation was applied rigorously to the Schwarzschild metric (3.5.1). The geodesics from this case were compared to the non-relativistic case, with a major difference being the presence of perihelion precession in the relativistic case. We also applied the geodesic equation to investigate the properties of the metric in de Sitter space (3.6.1). We found that energy is not conserved in de Sitter space, and that the universe increases in positive curvature with an increase in positive cosmological constant.

4 The Covariant Derivative

The main purpose of this section is to define and explore the covariant derivative. We start by defining the covariant derivative and acting it on vector fields and one-forms. With this result in mind, we explore the physical meaning of the covariant derivative through parallel transport (also known as the affine connection). We then understand some consequences originating from the path dependence of parallel transport and investigate geodesic deviation. To conclude this section, we will analyze how all the previous sections tie into the covariant derivative.

4.1 Defining the Covariant Derivative

We know that one way of moving differentiation to manifolds is through the lie derivative (see Appendix A). This type of differentiation gives sensible results for vector fields, one-forms and functions but fails to properly differentiate tensors (we get a mixed result). To tackle this issue, we introduce the covariant derivative. This object can be considered to be a "connection" which connects maps the distinct tangent spaces on a manifold together. We will first define this object mathematically, a more rigorous analysis of the connection will follow afterwards.
If $\mathcal{X}(M)$ is the set of all vector fields defined on a manifold, we define a connection to be a map $\nabla : \mathcal{X}(M) \times \mathcal{X}(M) \mapsto \mathcal{X}(M)$. This is usually written as: $\nabla(X, Y) = \nabla_X Y$, with ∇_X being called the covariant derivative. This object needs to satisfy the properties:

1. $\nabla_X(fY) = f\nabla_X(Y) + (\nabla_X f)Y$
2. $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$
3. $\nabla_{fX+gY}(Z) = f\nabla_X Z + g\nabla_X Z$

These conditions ensure that ∇ acts like an ordinary derivative on a scalar function, and exhibits similar mathematical properties as a partial derivative. We define, $\nabla_X f = X(f)$. We also write ∇ so that for a basis spanned by e_μ:

$$\nabla_{e_\rho} e_\mu = \Gamma^\nu_{\rho\mu} e_\nu \tag{4.1.1}$$

Here, $\Gamma^\nu_{\rho\mu}$ are the components of the connection. It is worth noting that these components are not tensors. This is because under a new system of coordinates, these components do not transform like a tensor. From now on, we use the notation: $\nabla_\mu = \nabla_{e_\mu}$. This makes typing this much easier and also increases resemblance to the partial derivative. It turns out that the covariant derivative is the best analog we have to the partial derivative on a manifold. To see why, consider its action on a vector field:

$$\nabla_X Y = \nabla_{X^\mu e_\mu}(Y^\nu e_\nu) = X^\mu \nabla_\mu (Y^\nu e^\nu) \tag{4.1.2}$$

Expanding out $\nabla_\mu(Y^\nu e^\nu)$ and using (4.1.1), this simplifies to:

$$\nabla_X Y = X^\mu (e_\nu(Y^\mu) + \Gamma^\nu_{\mu\rho} Y^\rho) e_\nu \tag{4.1.3}$$

We can get rid of the X^μ factor here. Essentially we can always write $\nabla_X = X^\mu \nabla_\mu$. This factor allows us to think of ∇ as a differential operator. Similarly we get:

$$\nabla_\mu Y = (e_\nu(Y^\mu) + \Gamma^\nu_{\mu\rho} Y^\rho) e_\nu \tag{4.1.4}$$

Now, this equation is already written in component form, so we can isolate the components of $\nabla_\mu Y$ as:

$$(\nabla_\mu Y)^\mu = e_\nu(Y^\mu) + \Gamma^\nu_{\mu\rho} Y^\rho = \nabla_\mu Y^\mu \tag{4.1.5}$$

We have introduced some new notation in this equation. Since we can write the covariant derivative as a differential operator, we can think of it as the natural generalization of the partial derivative. To further see this, assume that there is no curvature, which implies $\Gamma = 0$ (the relation to curvature can be seen later). This means that the second term in (4.1.4) becomes 0 and we essentially get the action the usual derivative would take.

This also allows us to physically decode the terms in (4.1.4) The first term can be visualized as the action of the usual derivative on Y^μ, this is just the rate of change of the component as we move along a curve. The second term on the other hand can be interpreted as the extra rate of change due to the curvature of our manifold M.

This way, we can also draw parallels from the covariant derivative to the substantial/total derivative seen in fluid dynamics and plasma physics. In this derivative, there is a term which accounts for the rate of change in a property due to the movement of the finite element along the field. This is in addition to one term accounting for the usual change in that property.

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \nabla.\mathbf{V} \tag{4.1.6}$$

The last task in this section is to act the covariant derivative on one-forms. When you know how to differentiate one-forms and vector fields, differentiating a tensor becomes easy (check Appendix A). For one-forms, we include in the definition of a covariant a fourth property:

$$\nabla_X(w(Y)) = (\nabla_X w)(Y) + w(\nabla_X Y) \tag{4.1.7}$$

This property nicely maintains the structure of the covariant derivative in that both left and right hand sides are functions. However, since these are functions in the end, we should also expect:

$$\nabla_X(w(Y)) = X(w(Y)) \tag{4.1.8}$$

Now, we can combine (4.1.7) and (4.1.8) while bring in the coordinate system to get:

$$X^\mu(\nabla_\mu w)_\nu Y^\nu = X^\mu(\partial_\mu w_\rho - \Gamma^\nu_{\mu\rho} w_\nu) Y^\rho \tag{4.1.9}$$

This allows us to isolate the components and introduce some new notation for one-forms:

$$(\nabla_\mu w)_\rho = \partial_\mu w_\rho - \Gamma^\nu_{\mu\rho} w_\nu = \nabla_\mu w_\rho \tag{4.1.10}$$

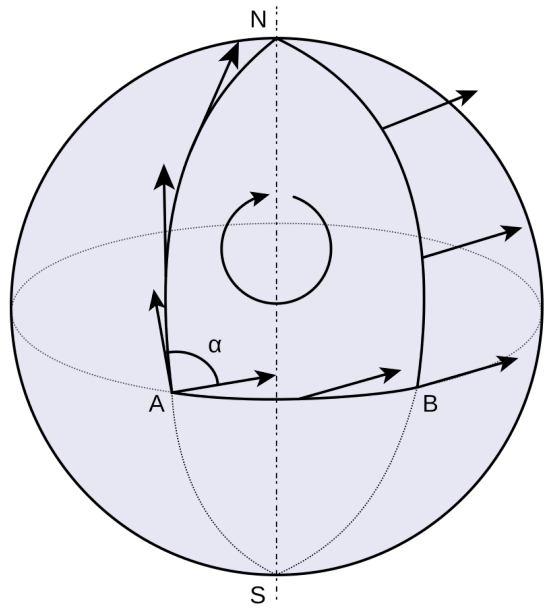

Figure 7: A great visualization of parallel transport (I borrowed this from Wikipedia)

Its nice to know that this result is the same as for the vector field but with a minus sign. In fact, we got the same pattern for the action of the lie derivative on a vector field and a one-form. It may be worth investigating into why this is. However, now that we know how to differentiate a one-form and a vector field, it is possible to differentiate tensors with a simple application of the results obtained in this section.

4.2 Parallel Transport

In the previous section, we touched upon the fact that the covariant derivative serves as a connection between tangent spaces. This connection allows us to introduce the concept of parallel transport. Here, all we are doing is pushing a vector along a curve on the manifold. (Imagine you go from the north to the south pole of the earth while trying your best to keep looking directly upwards) We can use a vector field X to define a curve C on our manifold. If the curve C is parameterized by co-ordinates $x^\mu(\tau)$, we define the vector field as:

$$X^\mu \mid_C = \frac{dx^\mu(\tau)}{d\tau} \tag{4.2.1}$$

A vector field Y is said to be parallely transported along C if it obeys:

$$\nabla_X Y = 0 \tag{4.2.2}$$

Expanding (4.2.2) in the coordinate system using (4.1.2) and $e_\mu = \partial_\mu$ gives us:

$$X^\nu(\partial_\nu Y^\mu + \Gamma^\mu_{\nu\rho} Y^\rho) = 0 \tag{4.2.3}$$

Expanding this gives us a series of coupled differential equations that allow us to generate the vector field at any point given an initial condition.

$$\frac{dY^\mu}{d\tau} + X^\nu \Gamma^\mu_{\nu\rho} Y^\rho = 0 \tag{4.2.4}$$

The presence of these equations is why the connection maps vector fields into vector fields and is called a "connection". Equation (4.2.4), if given a curve and connection as an input allows us to generate a new unique vector field (which implies the generation of a new vector at different points). The condition of $\nabla_X Y = 0$ also can be interpreted as telling us that the vectors given by Y don't change as we move along the curve. This is what allows us to interpret the new vectors generated using (4.2.4) as the same vector transported along the curve. Figure 7 is a great visualization of parallel transport. It shows how a vector changes after parallel transport along a curve in curved spacetime. We can see how the vector starts by pointing at one direction and ends up pointing in a completely different direction after returning back to its origin point. If we had transported the vector along a different curve, the final vector after transport would look different than it does now. This can also be extended to a tensor field, a tensor field is parallely transported if it obeys:

$$\nabla_X T = 0 \tag{4.2.5}$$

It is important to note that parallel transport is necessarily path-dependent.
The concept of parallel transport (also called the affine connection) is very useful with a few applications being defining normal coordinates (check Appendix B) and reframing the geodesic equation.

4.3 Revisiting the Geodesic Equation

Now that we have properly defined the covariant derivative and explored parallel transport, what happens if we parallel transport X along itself?
We can start by imposing the general condition:

$$\nabla_X X = X^\nu(\partial_\nu X^\mu + \Gamma^\mu_{\nu\rho} X^\rho) = \frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\tau}\frac{dx^\rho}{d\tau} \tag{4.3.1}$$

We have regained the geodesic equation here. This means that we can write the covariant geodesic equation simply as:

$$\nabla_X X = 0 \tag{4.3.2}$$

Since we can think of the covariant derivative ∇ as the analogue to the partial derivative, we can then think of the geodesics generated by (4.3.2) as a tangent curve to the curve generated by X.

For the Levi-Civita connection, we know that $\nabla_X g = 0$. If we parallely transport a vector field Y along the geodesic to the vector field X, we get the necessary condition:

$$\frac{dg(X,Y)}{d\tau} = 0 \tag{4.3.3}$$

This implies that the dot product of X and Y are constant as we move along the geodesic. This also implies that the angle between Y and X is constant along the curve. We can imagine this as Y curves alongside X along the geodesic (since the angle between them is constant).

4.4 Geodesic Deviation

We can investigate geodesics further. We write the geodesics using the coordinates $x^\mu(\tau; s)$. Here τ represents the affine parameter for the geodesic while s labels different geodesics. We know that the geodesics will be tangent to their vector field X. We write X as:

$$X^\mu = \frac{\partial x}{\partial \tau} \mid_s \tag{4.4.1}$$

Similarly to X, we define another vector field S to generate the "difference" between geodesics, parameterized by s. We write this as:

$$S^\mu = \frac{\partial x}{\partial s} \mid_\tau \tag{4.4.2}$$

We call the vector S^μ, the geodesic deviation vector. The name deviation vector is given as differentiating with respect to s essentially tells us how far apart two different geodesics are or alternatively, how "long" it would take to cross to another geodesic.
Since the whole family of geodesics spans a surface on the manifold, there shouldn't be any hinderance to choosing a coordinate system where $S = \partial/\partial s$ and $X = \partial/\partial\tau$ implying $[X, S] = 0$.
So far, we know that geodesics can deviate from each other with this deviation measured by the deviation vector S^μ. We would like to know more about how much these geodesics deviate as we move through τ. To get this equation, we start by assuming a torsion-free connection (This condition is not necessarily a requirement, its just more practical as general relativity mostly works with the Levi-Civita connection which is torsion free) (Check Appendix B for more on the Levi-Civita connection and Torsion). Torsion-free condition implies:

$$\nabla_X S - \nabla_S X = [X, S] = 0 \tag{4.4.3}$$

Projecting both sides with the covariant derivative will gives us:

$$\nabla_X \nabla_X S = \nabla_X \nabla_S X = \nabla_S \nabla_X X + R(X, S)X \tag{4.4.4}$$

Here, $R(X, Y)$ is the differential operator version of the Riemann tensor (Check Appendix B for more on the Riemann tensor). Since we are working with

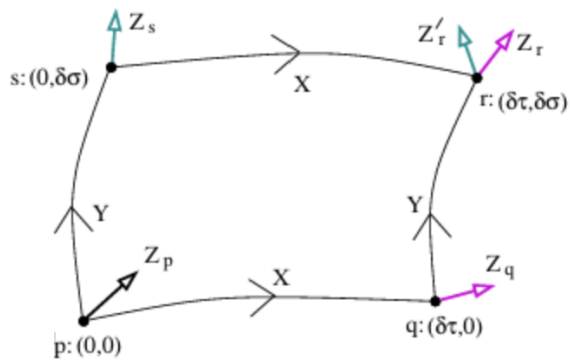

Figure 8: The setup for our analysis of path dependence.

geodesics, by definition $\nabla_X X = 0$. This means that the $\nabla_S \nabla_X X$ term goes to 0. Expanding for a coordinate system is:

$$X^\nu \nabla_\nu (X^\rho \nabla_\rho S^\mu) = R^\mu_{\nu\rho\sigma} X^\nu X^\rho S^\sigma \tag{4.4.5}$$

Restricting ourself to the curve C defined by the vector field X gives us:

$$\frac{D^2 S}{D\tau^2} = R^\mu_{\nu\rho\sigma} X^\nu X^\rho S^\sigma \tag{4.4.6}$$

Here, $D/D\tau = \partial x^\mu / \partial \tau \cdot \nabla_\mu$ is the covariant derivative for the curve C. This equation tells us that the relative acceleration for different geodesics depends on the Riemann curvature tensor. These geodesic deviations are called tidal forces. As we will see more in the next section, the Riemann tensor is actually very significant for parallel transport as it encodes geodesic deviation and path dependence.

4.5 Path Dependence

This explanation will use the same set-up as David Tong's lecture notes because its really nice. Let's start by looking at Figure 8. The question we are asking ourselves is, if we start with the same vector at point p, how will the final vector differ if we parallel transport it to point q using two different paths (shown in the figure) to get there.

We are using normal co-ordinates here, so the manifold is locally flat at point p. The usage of normal coordinates implies that moving from one point to another is just moving along a specific affine parameter. This translation is done using the vector fields $X = \partial/\partial\tau$ and $Y = \partial/\partial\sigma$. We are also assuming that all the points are in each others immediate neighborhood (this is simply to make our life simpler). The coordinates of each point are written on the figure.

Let's start by parallely transporting Z_p to Z_q along X. This needs to satisfy the parallel transport equation (4.2.2).

$$\frac{dZ^\mu}{d\tau} + X^\nu \Gamma^\mu_{\rho\nu} Z^\rho = 0 \tag{4.5.1}$$

However, since the manifold is locally flat at p, the Christoffel symbols vanish and so does the first derivative of Z^μ. We can Taylor expand Z^μ_q about the origin (p) as:

$$Z^\mu_q = Z^\mu_p + \frac{dZ^\mu}{d\tau}\ |_{\tau=0}\ \delta\tau + \frac{1}{2}\frac{d^2 Z^\mu}{d\tau^2}\delta\tau^2 + \mathcal{O}(\delta\tau^3) \tag{4.5.2}$$

We now need the second derivative, to get this we can differentiate (4.5.1) again and get:

$$\left.\frac{d^2 Z^\mu}{d\tau^2}\right|_{\tau=0} = -(X^\nu X^\sigma Z^\rho \Gamma^\mu_{\rho\nu,\sigma})_p \tag{4.5.3}$$

Since the understanding of path dependence and what it depends on is achieved by just the final result, I am not going to go into too much detail with the steps, all missing steps are given in the lecture notes. Thus we get Z^μ_q as:

$$Z^\mu_q = Z^\mu_p - \frac{1}{2}(X^\nu X^\sigma Z^\rho \Gamma^\mu_{\rho\nu,\sigma})_p \delta\tau^2 + \cdots \tag{4.5.4}$$

Notice how assuming the other points in the immediate neighborhood of p allows us to neglect the higher order terms from our calculation.
We can now parallel transport along Y to Z^μ_r. However, this time the parallel transport equation does not 0 out due to the normal co-ordinates. Doing similar shenanigans with the Taylor series gives us:

$$Z^\mu_r = Z^\mu_p - \frac{1}{2}(\Gamma^\mu_{\rho\nu,\sigma})_p [X^\nu X^\sigma Z^\rho \delta\tau^2 + 2Y^\nu Z^\rho X^\sigma \delta\sigma\delta\tau + Y^\nu Y^\sigma Z^\rho \delta\sigma^2]_p + \cdots \tag{4.5.5}$$

Travelling over the other path gives the same result but with X and Y swapped. This vector is denoted by Z'^μ_r
With this result, we can compare the vectors Z^μ_r and Z'^μ_r. This is done by:

$$\Delta Z^\mu_r = Z^\mu_r - Z'^\mu_r = -(\Gamma^\mu_{\rho\nu.\sigma} - \Gamma^\mu_{\rho\sigma,\nu})_p (Y^\nu Z^\rho X^\sigma)_p \delta\sigma\delta\tau + \cdots \tag{4.5.6}$$

Introducing the Riemann tensor allows us to simplify this equation to give us:

$$\Delta Z^\mu_r = (R^\mu_{\rho\sigma\nu} Y^\nu Z^\rho X^\sigma)_p \delta\sigma\delta\tau + \cdots \tag{4.5.7}$$

Since this whole equation is a tensor equation, the final result (4.5.7) is necessarily independent of coordinate systems. We can also tell from this equation that the Riemann tensor tells us about the path dependence of the parallel transport. We can also see how this is indicative of the curvature of the manifold. Curvature is ultimately what gives the covariant derivative path dependence. Since the Riemann tensor encodes the path dependence, it is not wrong to think it

also encodes the curvature of the manifold. This is why the Riemann tensor is also called the Riemann curvature tensor.

Using similar parallel transport arguments, we can also investigate the meaning of torsion. Lets say we have two vectors X and Y. We first parallel transport X along Y, this gives us a new point a. We then parallel transport Y along X which gives us a point b. Simply put, if the manifold is torsion free, this transport arrangement will give us $a = b$ and a closed parallelogram of vectors. However, if there is torsion in a manifold then the parallelogram will not close.

4.6 Interim Summary of the Project and some Physics

This whole section so far has been a mathematical analysis of the covariant derivative. This discussion still fails to answer why we need the covariant derivative for physics. To answer this question let's first look at what we learnt in the free particle section. This section took classical mechanics and made it relativistic through the use of Lagrangians. In the process we found the equation for a free particle moving through Minkowski space. In the next section on geodesics, we learnt that a geodesic can be viewed as a free particle moving through space. This definition implies that the free particle in Minkowski space is just a geodesic for that manifold. We then took this further and found an equation using Lagrangians that allowed us to calculate the geodesics for any manifold with a metric $g_{\mu\nu}(x)$. This was the geodesic equation (3.3.7). We also explored the immense physical significance of geodesics as the geodesic equation is what tells us how particles move in curved spacetime.

In this section, we tried to find the curve tangent to a vector field X in the direction of X. This action of the covariant derivative gave us the geodesic equation. Thus the most simple way of writing down the geodesic equation is due to the covariant derivative $\nabla_X X = 0$. In parallel with fields like aerodynamics and electrodynamics, we can think of geodesics as "streamlines" of the vector field X. It is a very well known fact that streamlines/field lines are very important in physics and this is no exception. This analysis also cements the covariant derivative ∇_X as the analogue of the partial derivative ∂ on a manifold M. All of these points underscore why the covariant derivative is will be a necessary object for our understanding of the physics of gravity.

In this section, we also learnt how geodesics deviate from each other due to the curvature of the manifold and how the covariant derivative is path dependent. These are some of the features of the covariant derivative and by extension curvature, that make it different from the partial derivative in R^n.

With all of the math established, we can quickly derive Einstein's equations for General Relativity before moving towards checking an application of the covariant derivative in QED.

5 Deriving Einstein's Equations

This section discusses Einstein's Equations. The first equation we derive is the equation in the absence of matter. This allows us to solve for any gravitational field. Using this equation, we quickly recap a few basic types of spacetime. We then investigate the symmetries of these spacetimes. This is followed by an analysis of the asymptotic values of these spacetimes. Finally, we look at how matter couples to the equations we already described.

5.1 Einstein's Equations in the Absence of Matter

A postulate of General Relativity is that the gravitational field is described by a metric $g_{\mu\nu}$ on a 4-D Lorentzian manifold.
Generally in physics, we create an action and then minimize it to get an equation of motion. We already know that the metric is a natural volume form for the manifold, so we really just need a function on this manifold to create the action.

5.1.1 Varying the Einstein-Hilbert Action

The simplest such function is given by the Ricci scalar R. Note that this function has a second order-derivative for the metric as $R \approx \partial\Gamma + \Gamma\Gamma$. This corresponds to most of the actions we see in physics which have a second order derivative of the field. Thus we have constructed what is known as the Einstein-Hilbert action.

$$S = \int d^4x\sqrt{-g}R \tag{5.1.1}$$

The task now is to minimize this action. If we make the transformation $g_{\mu\nu}(x) \to g_{\mu\nu}(x) + \delta g_{\mu\nu}(x)$, the change in the action is given by:

$$\delta S = \int d^4x((\delta\sqrt{-g})g^{\mu\nu}R_{\mu\nu} + \sqrt{-g}(\delta g^{\mu\nu})R_{\mu\nu} + \sqrt{-g}\ g^{\mu\nu}(\delta R_{\mu\nu})) \tag{5.1.2}$$

To simplify this, let's start by finding $\delta g_{\mu\nu}$. This can be found by varying the inverse metric equation $g_{\rho\mu}g^{\mu\nu} = \delta^{\nu}_{\rho}$. This gives us:

$$(\delta g_{\rho\mu})g^{\mu\nu} + g_{\rho\mu}\delta g^{\mu\nu} \Rightarrow \delta g^{\mu\nu} = -g^{\mu\rho}g^{\nu\sigma}\delta g_{\rho\sigma} \tag{5.1.3}$$

Using this we can find $\delta\sqrt{-g}$. This is done by applying the identity $log\ det(A) = tr(log\ A)$. An application of the identity gives us:

$$\delta\sqrt{-g} = \frac{1}{2}\frac{1}{\sqrt{-g}}(-g)g^{\mu\nu}\delta g_{\mu\nu} = \frac{1}{2}\sqrt{-g}\ g^{\mu\nu}\delta g_{\mu\nu} \tag{5.1.4}$$

Using the result from (5.1.3), we can write:

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}\ g_{\mu\nu}\delta g^{\mu\nu} \tag{5.1.5}$$

Now, we are only left with finding the variation of the Ricci tensor and putting

it all together. We will start by varying the Christoffel symbol and working in normal coordinates.

$$\delta\Gamma^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu\delta g_{\mu\nu} + \partial_\nu\delta g_{\sigma\mu} - \partial_\sigma\delta g_{\mu\nu}) = \frac{1}{2}g^{\rho\sigma}(\nabla_\mu\delta g_{\sigma\nu} + \nabla_\nu\delta g_{\sigma\mu} - \nabla_\sigma\delta g_{\mu\nu}) \tag{5.1.6}$$

Note that while Γ is not a tensor, $\delta\Gamma$ is a tensor and thus independent of coordinate changes. With the variation of the Christoffel symbol in hand, we can vary the Riemann tensor in normal coordinates ($R^{\sigma}_{\rho\mu\nu} = \partial_\mu\Gamma^{\sigma}_{\nu\rho} - \partial_\nu\Gamma^{\sigma}_{\mu\rho}$):

$$\delta R^{\sigma}_{\rho\mu\nu} = \partial_\mu\delta\Gamma^{\sigma}_{\nu\rho} - \partial_\nu\delta\Gamma^{\sigma}_{\mu\rho} = \nabla_\mu\delta\Gamma^{\sigma}_{\nu\rho} - \nabla_\nu\delta\Gamma^{\sigma}_{\mu\rho} \tag{5.1.7}$$

We can now contract the indices to get the final change in the Ricci tensor:

$$\delta R_{\rho\nu} = \nabla_\mu\delta\Gamma^{\mu}_{\nu\rho} - \nabla_\nu\delta\Gamma^{\mu}_{\rho\mu} \tag{5.1.8}$$

If we define $X^\mu = g^{\rho\nu}\delta\Gamma^{\mu}_{\rho\nu} - g^{\mu\nu}\delta\Gamma^{\rho}_{\nu\rho}$, we can finally put all the variations together to give us:

$$\delta S = \int d^4x\sqrt{-g}\left[\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right)\delta g^{\mu\nu} + \nabla_\mu X^\mu\right] \tag{5.1.9}$$

Now, the second term in the brackets is a derivative and we can apply the divergence theorem. Its worth noting that we get an analogous derivative even while deriving the equations of motion in classical field theory. This proves that we are on the right track. We can thus ignore that term and the necessary $\delta S = 0$ condition gives us:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = G_{\mu\nu} = 0 \tag{5.1.10}$$

Here, we have identified the Einstein tensor (defined in Appendix B). If we contract (5.1.10) with the inverse metric $g^{\mu\nu}$, we get an equation that is purely the Ricci scalar. To satisfy this equation, we require $R = 0$. This gives us the final equation as:

$$R_{\mu\nu} = 0 \tag{5.1.11}$$

These are the Einstein equations in the absence of matter. We just require the metric to be Ricci flat.

Its also worth noting that the Einstein-Hilbert action isn't dimensionally accurate. We require an action to be $[S] = L^2$. To get such an action we can multiply by a constant with the right dimensions to get the final Einstein-Hilbert action.

$$S = \frac{c^3}{16\pi G}\int d^4x\sqrt{-g}\ R \tag{5.1.12}$$

We will proceed further only in the units where $c = 1$, this makes life much easier.

5.1.2 Einstein's Equation with the Cosmological Constant

We could also have added a constant to the Einstein-Hilbert action. This action can look like:

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}(R - 2\Lambda) \tag{5.1.13}$$

Here, Λ is called the cosmological constant. It has the dimension $[\Lambda] = L^{-2}$. We can think of the cosmological constant as a "potential energy". The minus sign is there to motivate the $T - V$ interpretation of the action.
In the presence of the cosmological constant, the Einstein Equation becomes:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = -\Lambda g_{\mu\nu} \tag{5.1.14}$$

Contracting the equation with the inverse metric tells us that $R = 4\Lambda$. This gives us the equation:

$$R_{\mu\nu} = \Lambda g_{\mu\nu} \tag{5.1.15}$$

5.1.3 What have we actually derived?

Before we answer this question, a quick note from quantum field theory. The process of constructing a theory to describe a scalar field starts by defining the field itself. The scalar field is derived via the action:

$$S = \int d^4x \left(\frac{1}{2}\partial^\mu\phi\partial_\mu\phi - \frac{1}{2}m^2\phi^2\right) \tag{5.1.16}$$

The equation of motion we get by varying this action is given by:

$$(\partial^2 - m^2)\phi^2 = 0 \tag{5.1.17}$$

This is the Klein-Gordon equation. Now, the field we get from this equation doesn't describe much fun physics but it is the cornerstone of spin-zero quantum field theories.
This is exactly what we have done here, Einstein's Equations can be thought of as describing the base field. Metrics such as the Schwarzschild metric can be verified to satisfy Einstein's Equation:

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right)dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1}dr^2 + r^2(d\theta^2 + sin^2\theta\ d\phi^2) \tag{5.1.18}$$

We now examine a few basic solutions of (5.1.15). These can be thought of as simply describing spacetime without any planets or stars.

5.1.4 Different Spacetimes

There are three types of spacetime that we are mostly concerned about. These are Minkowski, de Sitter (dS) and Anti de Sitter (AdS) spacetimes. These spacetimes are given by the solution of (5.1.15) when $\Lambda = 0$, $\Lambda > 0$ and $\Lambda < 0$ respectively.

We have already investigated the geodesics in Minkowski and de Sitter space. The analysis of AdS spacetime is essentially the same. It is also worth noting that AdS spacetime makes a particle oscillate about a point, in contrast to dS spacetime which tends to push a particle further out.

5.1.5 An Aside: Embedding Spacetimes

Now, we know that its important to not think of our 4D Lorentzian manifold as embedded in any higher dimensional space. This is the right way to think about general relativity. However, nothing really prevents us from embedding a spacetime into a higher dimensional space just to get a better understanding of its properties.
For example, (without going into the math) we can embed de Sitter space into a 5D Minkowski space. Following the math tells us that this spacetime acts like a hyperboloid in 5D space which tells us that the universe is expanding.

5.2 Symmetries of Spacetime

Symmetries are very important in General Relativity. The basic equations we need to solve for a gravitational field are $R_{\mu\nu} = 0$. Here, there are no visible parameters to tweak. Parameters are usually present in partial differential equations and tweaking them gives us the results for different configurations.
The question is, how do we differentiate different configurations for Einstein's equations without these parameters? One way to do this is by looking at symmetries. This outlook can be explained using the Schwarzschild solution which describes any spherically symmetric object. Notice how it applies to any spherically symmetric body and not just a star or a blackhole. This characteristic of the Schwarzschild solution makes the importance of symmetry clear.
We can now investigate symmetries in General Relativity in more detail.

5.2.1 Killing Vectors

Let's start with a flow defined on a manifold M. This flow is generated by a vector field $K \in \mathcal{X}(M)$. We know that the vector field is tangent to each point along the flow. If we express the flow in coordinates $x^\mu(t)$ we have:

$$K^\mu = \frac{dx^\mu}{dt} \tag{5.2.1}$$

Now, we get to the symmetry bit. If a metric looks the same along the curve generated by the flow, the flow is called an isometry. Mathematically, this condition is given by:

$$\mathcal{L}_K g = 0 \Leftrightarrow \nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \tag{5.2.2}$$

This is known as the Killing equation. If a metric does not depend on a certain coordinate (say x^i) then we can easily find a killing vector $K = \frac{\partial}{\partial x^i}$. This fact is important because we know that if a metric does not depend on a certain coordinate, the action won't depend on it either and if the action doesn't depend on a certain coordinate it implies a conserved quantity. This illustrates a link

between Killing vectors and conservation laws.

There is another aspect about Killing vectors that we need to discuss. Using the result $\mathcal{L}_X\mathcal{L}_Y - \mathcal{L}_Y\mathcal{L}_X = \mathcal{L}_{[X,Y]}$, we can conclude that Killing vectors form a Lie Algebra if X, Y are Killing vectors. This is said to be the Lie algebra of the isometry group of the manifold.

5.2.2 Quick Detour into Minkowski Space

The killing equation for Minkowski space is given by:

$$\partial_\mu K_\nu + \partial_\nu K_\mu = 0 \tag{5.2.3}$$

The solutions to this can be of two forms:

$$K_\mu = c_\mu \ ; K_\mu = w_{\mu\nu}x^\nu \tag{5.2.4}$$

Here, $w_{\mu\nu}$ has to be anti-symmetric to satisfy the Killing equation.
With the general form of the components of Killing vectors in hand, we can define a few killing vectors:

$$P_\mu = \partial_\mu \ ; M_{\mu\nu} = \eta_{\mu\nu}x^\rho\partial_\nu - \eta_{\nu\rho}x^\rho\partial_\mu \tag{5.2.5}$$

There are 10 Killing vectors in total and they are very familiar. These vectors are the generators or translations and rotations. These non-zero commutators are given by:

$$[M_{\mu\nu}, P_\sigma] = -\eta_{\mu\sigma}P_\nu + \eta_{\sigma\nu}P_\mu \tag{5.2.6}$$

$$[M_{\mu\nu}, M_{\rho\sigma}] = \eta_{\mu\sigma}M_{\nu\rho} + \eta_{\nu\rho}M_{\mu\sigma} - \eta_{\mu\rho}M_{\nu\sigma} - \eta_{\nu\sigma}M_{\mu\rho} \tag{5.2.7}$$

Including in $[P_\mu, P_\nu] = 0$ allows us to identify the Lie algebra of the Poincare group $\mathbf{R}^4 \times SO(1,3)$. This example hits home the fact that each isometry has its own isometry group associated with it.
In fact, if we repeat this analysis on de Sitter and anti de Sitter spaces, we find that they are invariant under $SO(1,4)$ and $SO(2,3)$ isometry group respectively. This analysis can be done by embedding into a 4D spacetime. Through this method, we also find that AdS has a Killing vector that is timelike ($g_{\mu\nu}K^\mu K^\nu < 0$) everywhere but that is not true for de Sitter space.
The timelike nature of a Killing vector is an important property which we will investigate later.

5.3 Asymptotics of Spacetime

If we are given a metric $g_{\mu\nu}$ that is not visibly de Sitter, anti de Sitter or Minkowski. How do we identify which type of spacetime this metric describes? To answer this question we need to go to infinity in the spacetime described by $g_{\mu\nu}$. The three types of spacetime have different behaviours at infinity and by analyzing these behaviours we can answer our question.
The first step in our analysis is to look at conformal transformations.

5.3.1 Conformal Transformations

If we have a spacetime M dscribed by a metric $g_{\mu\nu}$, we can construct a new metric $\bar{g}_{\mu\nu}$ by a conformal transformation.

$$\bar{g}_{\mu\nu}(x) = \Omega^2(x) g_{\mu\nu}(x) \tag{5.3.1}$$

Here, $\Omega(x)$ is a smooth, non-vanishing function. This transformation serves to warp the distances between spacetimes. However, it does not change the angles. For example, if we have $g_{\mu\nu}X^\mu X^\nu = 0$, so the vector field X is globally null w.r.t $g_{\mu\nu}$ the conformal transformation will yield:

$$\bar{g}_{\mu\nu}X^\mu X^\nu = \Omega^2(x) g_{\mu\nu}X^\mu X^\nu = 0 \tag{5.3.2}$$

Since the conformal transformation preserves the angles (Causal structure), we can use it to warp distances and bring the infinity to us while still conserving the causal properties of spacetime.

Now, the Riemann tensor (and by extension the Ricci tensor) is not invariant under a conformal transformation. To fix this we define a new tensor termed the Weyl tensor. If our manifold has a dimension n, we have :

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{2}{n-2}(g_{\mu[\rho}R_{\sigma]\nu} - g_{\nu[\rho}R_{\sigma]\mu}) + \frac{2}{(n-1)(n-2)} R g_{\mu[\rho}g_{\sigma]\nu} \tag{5.3.3}$$

This tensor retains the symmetric properties of the Riemann tensor.

The invariance of the Riemann tensor is also indicative of another fact. Consider (4.4.6), we can see that since the Riemann tensor is not invariant to conformal transformations, geodesic should not be able to map to each other in between these two types of spacetime. It is possible to see that timelike geodesics do not map between each other. While timelike geodesics can't map to each other, null geodesics do map to each other with some issues with the affine parameterization.

5.3.2 Penrose Diagrams

We have learnt that we can theoretically use conformal transformations to "pull" the asymptotes of spacetime towards us. This is possible because the distances are warped under such transformations but angles and the causal structure is conserved. The causal structure that is the byproduct of a conformal transformation can be visualized using a Penrose diagram.

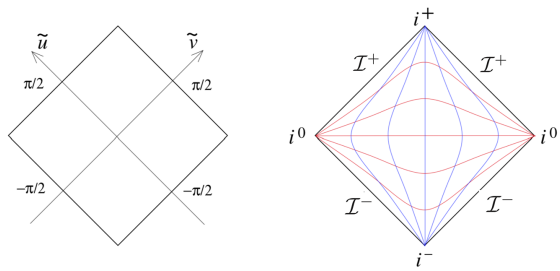

Figure 9: Penrose Diagrams for Minkowski Space

5.3.3 Conformal Transformations of Minkowski Space

Consider Minkowski space in $d = 1+1$ dimensions. Here the metric is given by:

$$ds^2 = -dt^2 + dx^2 \tag{5.3.4}$$

We start the process by switching coordinates to:

$$u = t - x \ ; \ v = t + x \Leftrightarrow ds^2 = -du \ dv \tag{5.3.5}$$

The problem is that these coordinates still extend to infinity. We can make the distance finite by changing coordinates to:

$$u = tan\bar{u} \ ; \ v = tan\bar{v} \Leftrightarrow ds^2 = -\frac{1}{cos^2\bar{u} \ cos^2\bar{v}} d\bar{u} \ d\bar{v} \tag{5.3.6}$$

We can now cover all of Minkowski space with just $\bar{u}, \bar{v} \in (-\frac{\pi}{2}, \frac{\pi}{2})$. We can't include the boundary points of $\pm\pi/2$ as this point makes the metric ill-defined. However, with the metric in this form, we can make a simply conformal transformation to give us a more workable metric.

$$d\bar{s}^2 = (cos^2\bar{u} \ cos^2\bar{v}) \ ds^2 = -d\bar{u} \ d\bar{v} \tag{5.3.7}$$

We can now include the boundary points again. This leaves us with a compact Minkowski space that we can represent in a Penrose diagram. The process that we just did is known as conformal compactification. The resulting Penrose diagrams are shown in Figure 9.
The convention is that light-rays are always at 45°. This means that the coordinate axes for $\bar{u}, \bar{v}$ will be drawn as shown in the right side of Figure 9. The left side of the figure shows the geodesics with respect to the original metric. The blue lines are time-like geodesics with constant x and the red lines are space-like geodesics with constant t. The information we obtain from the Penrose diagram is the following:

1. All timelike geodesics start at i^- and end at i^+. This is the origin and end-point of all massive particles. These points are known as the past and future timelike infinity respectively.

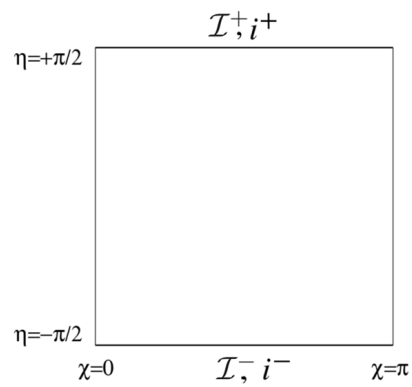

Figure 10: Basic Penrose diagram for de Sitter space.

2. All spacelike geodesics begin or end at one of the two points i^0. These points are known as spacelike infinity.

3. All null geodesics start at the boundary $\mathcal{I}^-$ and end at $\mathcal{I}^+$. These boundaries are known as past null infinity and future null infinity respectively.

We can see that there are more ways to go to infinity for null geodesics than its other counterparts. This is a characteristic feature of Minkowski space. We can also see that an observer in Minkowski space can "see everything" if they wait long enough. Also, given any two points in Minkowski space, their past and future light cones necessarily intersect (To see this, just draw light cones for any arbitrary two points on the diagram in Figure 9). This means that they are causally connected in both the past and the future. This also means that there was always an event in the past that could influence both points and one in the future that can influence both points.

5.3.4 Penrose Diagrams for de Sitter Space

For this section, we will skip the conformal compactification process and look straight at the Penrose diagram. Figure 10 shows the basic penrose diagram after conformal compactification. It is important to note that the vertical lines are not boundaries of spacetime. They are the poles of the $\mathbf{S}^3$ that is encoded into this Penrose diagram. Since this is not the diagram of 2D de Sitter space, each point on the Penrose diagram encodes a 2-sphere.
The horizontal lines at the top and bottom are the origin points of both the null and timelike geodesics. This is in contrast to Minkowski space where timelike geodesics could only originate from one point.
The feature of Minkowski space where if an observer waits long enough they could see all of spacetime is also not valid in de Sitter space. Consider an observer on the left vertical line (the north pole). Here, the observer will only be able to interact with / see half of spacetime. Figure 11 helps visualize the

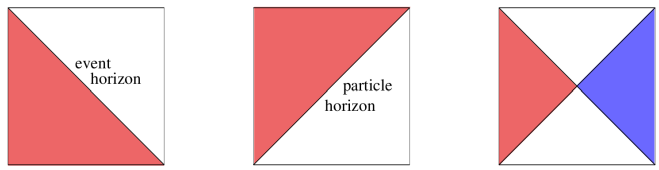

Figure 11: The causal structure of de Sitter space

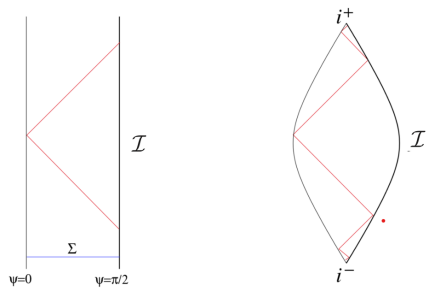

Figure 12: Penrose Diagram for Anti de Sitter Space.

causal structure of de Sitter space. The event horizon is the spacetime an observer can see if they wait at the north pole for "long enough". The particle horizon is the amount of spacetime the observer can interact with if they start at the beginning. It is also important to note that these horizons are observer dependent. For this reason, they are usually termed cosmological horizons. The intersection of these two horizons is termed the causal diamond. The intersection of the north pole causal diamond and south pole causal diamond is shown in the right of Figure 11.

The most important thing for us to keep in mind is that no single observer in de Sitter space has a full view of the spacetime. Physics in de Sitter space is observer dependent. This leads to various conceptual issues while studying topics such as Quantum Gravity.

5.3.5 Penrose Diagrams for Anti de Sitter Space

Similar to the previous section, we will skip the conformal compactification process for AdS space and move straight to the results. AdS space is easier to understand conceptually than de Sitter space. First off, we will consider the case where the time coordinate is infinite. The resulting Penrose diagram shown in the left side of Figure 12 is an infinite strip. The boundary $\mathcal{I}$ should be viewed as a combination of $\mathcal{I}^-, \mathcal{I}^+, i^0$. Since its normal vector is spacelike, we can view the boundary as timelike.

We can see that light rays hit the boundary in finite conformal time. This means that to describe any physics in AdS, we need to specify what happens at the boundary. The figure shows a light ray starting from a point and terminating after touching the boundary again. However, there is nothing stopping us from making the light ray bounce around the boundaries forever. This phenomenon can also be viewed as us giving a spacelike hypersurface $\sum$ some initial data and time-evolving it. However, we need to specify what happens on the boundary since there are other points in the future of $\sum$ which are in causal contact with the boundary.
Since there is no Cauchy surface on which we can specify initial data, we say that AdS is not globally hyperbolic.

The left side of Figure 12 shows a totally conformally compactified Penrose diagram of AdS. This also demonstrates the same conceptual properties as the previous infinite strip Penrose diagram. We can see that $\mathcal{I}$ is a timelike boundary from which a light ray can originate. Light rays also bounce around infinite times in this diagram. Additionally, we can also see a future and past timelike infinite $i^{\pm}$. This is where timelike particles originate/die.

5.3.6 A Quick Summary

This section introduced the idea of conformal transformations to us. Using conformal transformations, we were able to bring the asymptote of the spacetime to us for analysis. We also saw that we can visualize the conformally compactified spacetime using Penrose diagrams, which allows us to investigate the causal structures of spacetime.
We applied this process to investigate the causal structures of Minkowski, de Sitter and anti de Sitter spaces. We found that in Minkowski space, an observer can (in principle) wait long enough and see all of spacetime. This characteristic does not hold in de Sitter space which is observer dependent. Here, an observer can only interact and see a certain amount of spacetime. We also saw that the characteristic property of AdS space is that it makes massless particles bounce around endlessly (depending upon our chosen boundary condition) and traps massive particles.
Bringing this discussion back to the real world, de Sitter spacetime serves as the most likely candidate to describe our own universe. However, due to its observer dependent nature, it is not used to investigate our theories of quantum gravity. AdS is used to investigate the effects of quantum gravity. This choice is due to a property called AdS/CFT correspondence which will not be discussed in this document.

5.4 Matter and the Gravitational Field

So far, we have discussed Einstein's equations in the absence of matter. This picture is interesting and quite fun to study but it is not complete. Adding matter gives rise to a few changes to our equations. This is because matter interacts with the gravitational field instead of just existing. But before we

have a look at how matter impacts the gravitational field, we first need to see how the gravitational field impacts matter.
In this section, we will study the impact on a matter field as this is a pretty simple case to analyze.

5.4.1 Impact of $g_{\mu\nu}$ on Matter Fields ϕ

We know that the action for a scalar field $\phi(x)$ in Minkowski space looks like:

$$S_{scalar} = \int d^4x \left(-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi) \right) \tag{5.4.1}$$

This can be easily generalized to curved spacetime by swapping the metric and integrating over the curved spacetime volume form.

$$S_{scalar} = \int d^4x\sqrt{-g}\ \left(-\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(\phi) \right) \tag{5.4.2}$$

Note that we have swapped the partial derivative with the covariant derivative here since they are equivalent in this case ($\partial_\mu\phi = \nabla_\mu\phi$). We can also add an interaction term of the form $(1/2)\xi R\phi^2$ where ξ is the coupling parameter for the interaction.
This is analogous to how we add interactions to the base theory in quantum field theory. If we vary this interacting theory with respect to ϕ we get:

$$\delta S_{scalar} = \int d^4x\sqrt{-g}\left[\left(g^{\mu\nu}\nabla_\mu\nabla_\nu\phi - \frac{\partial V}{\partial\phi} - \xi R\phi \right)\delta\phi - \nabla_\mu(\delta\phi\nabla^\mu\phi) \right] \tag{5.4.3}$$

The final term is a boundary term and we can forget about it. This gives us the equation of motion for matter in curved spacetime:

$$g^{\mu\nu}\nabla_\mu\nabla_\nu\phi - \frac{\partial V}{\partial\phi} - \xi R\phi = 0 \tag{5.4.4}$$

Now that we know how matter reacts to curved spacetime, we can look at how curved spacetime (gravitational field) reacts to matter.

5.4.2 The Einstein Equations with Matter

To analyze this situation, we need to look at the combined action:

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}(R - 2\Lambda) + S_M \tag{5.4.5}$$

Here S_M denotes the action for matter fields. This allows for any possible interaction terms in S_M and is thus a general matter action. Since it is a general action, we need to define its variation with respect to $g_{\mu\nu}$ to proceed any further. This definition is known as the energy-momentum tensor.

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}} \tag{5.4.6}$$

Using this definition, the variation of the combined action S with respect to the gravitational field $g_{\mu\nu}$ is:

$$\delta S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,\left(G_{\mu\nu} + \Lambda g_{\mu\nu}\right)\,\delta g^{\mu\nu} - \frac{1}{2}\int d^4x\sqrt{-g}\;T_{\mu\nu}\;\delta g^{\mu\nu} \tag{5.4.7}$$

This gives us the full Einstein equations which describe gravity coupled to matter.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{5.4.8}$$

It is convention to take the cosmological constant as a component of the energy momentum tensor.

$$(T_{\mu\nu})_\Lambda = -\frac{\Lambda}{8\pi G} g_{\mu\nu} \tag{5.4.9}$$

This allows us to write the Einstein equations in a more compact form.

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{5.4.10}$$

Where $T_{\mu\nu}$ includes the cosmological term. If we switch this equation from being about the Ricci tensor R instead of the Einstein tensor G. This gives us the alternative form of Einstein's equation.

$$R_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{1}{2}T g_{\mu\nu}\right) \tag{5.4.11}$$

5.4.3 A Note on the Energy-Momentum Tensor

By considering diffeomorphism invariance, we can find that the energy-momentum tensor is necessarily covariantly conserved.

$$\nabla_\mu T^{\mu\nu} = 0 \tag{5.4.12}$$

Note that covariant conservation is not necessarily actual conservation. Energy and momentum conservation in general relativity is very subtle and requires a whole analysis of itself. This won't be covered in this document.

6 The Covariant Derivative in U(1)

This section discusses the significance of $U(1)$ symmetry in both non-relativistic and relativistic theories of the electromagnetic interaction. We then define a new derivative, which we will also call the covariant derivative that accounts for the changes made to the equation to make it invariant to the $U(1)$ symmetry, also known as a gauge symmetry.

6.1 Schrodinger's Equation and Gauge Symmetry

To understand how the covariant derivative applies to quantum mechanics and quantum field theory, we first need to investigate Schrodinger's equation and its relation to the $U(1)$ symmetry. The $U(1)$ transformations look like:

$$\Psi \mapsto e^{-i\alpha}\Psi \tag{6.1.1}$$

This is also referred to as a gauge transformation. If α is not a function of x, t then we have what is known as a global gauge transformation, otherwise it is known as a local gauge transformation. Now, looking at Schrodinger's equation for a free particle we have:

$$-\frac{1}{2m}\nabla^2\Psi(\mathbf{x},t) = \frac{\partial}{\partial t}\Psi(\mathbf{x},t) \tag{6.1.2}$$

Making the $U(1)$ transformation.

$$-\frac{1}{2m}\left((\nabla^2 e^{-i\alpha})\Psi(\mathbf{x},t) + e^{-i\alpha}(\nabla^2\Psi(\mathbf{x},t)\right) = \frac{\partial e^{-i\alpha}}{\partial t}\Psi(\mathbf{x},t) + e^{-i\alpha}\frac{\partial}{\partial t}\Psi(\mathbf{x},t) \tag{6.1.3}$$

It is clear that this equation is not invariant to a local gauge transformation. So, how do we make an invariant equation? Consider the equation:

$$\frac{1}{2m}(-i\nabla + e\mathbf{A})^2\Psi(\mathbf{x},t) = \left(i\frac{\partial}{\partial t} + eV\right)\Psi(\mathbf{x},t) \tag{6.1.4}$$

We have inserted the arbitrary fields $\mathbf{A}$ and V. This is a tool that we can use to eliminate the extra terms in Eq.(6.1.3). So if we make the transformation:

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \frac{1}{e}\nabla\alpha \ ; \ V \mapsto V' = V - \frac{1}{e}\frac{\partial\alpha}{\partial t} \tag{6.1.5}$$

We now have a modified Schrodinger's equation that is invariant to $U(1)$ symmetry. Happily enough, Eq. (6.1.4) has a physical meaning. It is the equation for an electrically charged particle, the electron. The vector field $A^\mu = (V, \mathbf{A})$ is associated with a spin-1 particle. This can be interpreted to be a photon. This means that physically, the electromagnetic interaction and the existence of the photon are a feature of $U(1)$ invariance.

6.2 Aside: U(1) Symmetries for Quantum Electrodynamics

In Quantum Field Theory, the electron field is described by a $(2,1)$, spinor representation of the Lorentz Group. The simplest such representation is given by the Weyl field given by ψ. The Lagrangian for a Weyl theory is given by:

$$\mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2} m\psi\psi - \frac{1}{2} m^* \psi^\dagger \psi^\dagger \tag{6.2.1}$$

Here, all Spinor indices are suppressed. We want to impose $U(1)$ invariance on this Weyl Theory now. To do this, we actually start by creating a theory that is invariant to $SO(2)$ since $SO(2)$ maps into $U(1)$ directly. This invariance can be imposed by adding two Weyl theories together. This gives us an $SO(2)$ invariant theory.

$$\mathcal{L} = i\psi_i^\dagger \bar{\sigma}^\mu \partial_\mu \psi_i - \frac{1}{2} m\psi_i\psi_i - \frac{1}{2} m\psi_i^\dagger \psi_i^\dagger \tag{6.2.2}$$

We can now write the two Weyl fields in terms of a few new fields:

$$\chi = \frac{1}{\sqrt{2}}(\psi_1 + i\psi_2) \ ; \ \xi = \frac{1}{\sqrt{2}}(\psi_1 - i\psi_2) \tag{6.2.3}$$

We can now define the Dirac field:

$$\Psi = \begin{pmatrix} \chi_c \\ \xi^{\dagger\dot{c}} \end{pmatrix} \tag{6.2.4}$$

(Here the spinor indices are no longer suppressed as they are important for the definition of the Dirac field) This gives us the Lagrangian for the "Dirac" field.

$$\mathcal{L} = i\bar{\Psi}\gamma^\mu \partial_\mu \Psi - m\bar{\Psi}\Psi \tag{6.2.5}$$

This theory is invariant under $U(1)$ symmetry. This $U(1)$ symmetry gives rise to a Noether current which we can identify as the electromagnetic current in Quantum Electrodynamics. This means that even from the QED perspective, $U(1)$ symmetry gives us a theory of electromagnetic interactions.

6.3 The Covariant Derivative for QED

If we now define the covariant derivative:

$$\mathcal{D}^\mu = \partial^\mu - ieA^\mu \tag{6.3.1}$$

We can rewrite the electromagnetic Schrodinger equation in the free particle form as:

$$\frac{1}{2m}(i\mathbf{D}^2)\Psi = i\mathcal{D}^0\Psi \tag{6.3.2}$$

It is easy to check that this theory is invariant to the gauge transformations given in Eq. (6.1.5). This tells us that the covariant derivative here acts as a derivative which accounts for gauge theory based changes in the wave function. This similar to how the differential geometry covariant derivative accounts for curvature.

7 Appendix A: Differential Geometry

This appendix discusses a few topics from differential geometry that find their use in General Relativity. This discussion is not meant to be rigorous, its only purpose is to be a quick summary and to improve my own understanding of the concepts involved.

7.1 Manifolds

Manifolds form the base of all relevant mathematics. It is useful to think of these as a space we can freely deform and mess around with. Formally, we can define a differentiable manifold as:
A Manifold M is a Haussdorf Topology T which:

1. is locally homeomorphic to R^n. For each point $p \in M$, we have a homeomorphism: $\phi : \mathcal{O} \mapsto \mathcal{U}$, where $\mathcal{O} \in M$ is an open-subset that is the neighbourhood of ϕ and $\mathcal{U}$ is an open-subset of R^n

2. Has two compatible maps between corresponding to open-subsets $\mathcal{O}_\alpha$ and $\mathcal{O}_\beta$. This means that for maps: $\phi_\alpha : \mathcal{O}_\alpha \mapsto \mathcal{U}_\alpha$ and $\phi_\beta : \mathcal{O}_\beta \mapsto \mathcal{U}_\beta$, The map $\phi_\beta \circ \phi_\alpha^{-1}$ is C^∞.

We call the maps ϕ_α charts and a collection of all these charts as an atlas. We can think of ϕ_α as one coordinate system, ϕ_β as another. The compatibility condition ensures that there are no inconsistencies between the two co-ordinate systems. (Basically, the same point $p \in M$ should be represented accurately in R^n by both charts) // A function $f : \mathcal{O} \mapsto R$ defined on a manifold M is said to be C^∞ if $f \circ \phi^{-1} : \mathcal{U} \mapsto R$ is C^∞.
We can define the notion of similarity for maps between manifolds in a similar fashion.
A diffeomorphism is a C^∞ homeomorphism between two manifolds, $f : M \mapsto N$. The two manifolds M, N are said to be diffeomorphic and they must share the same dimension to ensure the existence of an inverse.

7.2 Tangent Spaces

If the geometry is curved, the notion of a vector as a straight arrow is no longer valid. The solution to this problem is to define a tangent space for each point $p \in M$. This will allow us to create vectors at each point without worrying about co-ordinate systems. We define the tangent vector X_p (which is an object that differentiates C^∞ functions at a point $p \in M$), $X_p : C^\infty(M) \mapsto R$

1. X_p should be linear: $X_p(f + g) = X_p(f) + X_p(g)$ for all $f, g \in C^\infty(M)$

2. X_p should exhibit Leibnizarity: $X_p(fg) = f(p)X_p(g) + X_p(f)g(p)$ for all $f, g \in C^\infty(M)$

3. $X_p(c) = 0$ where c is a constant function

Let $F = f \circ \phi^{-1} : \mathcal{U} \mapsto R$ be the function we are currently concerned about. We can write:

$$F(x) = F(x^\mu(p)) + (x^\mu - x^\mu(p))F_\mu(x) \Rightarrow \frac{\partial F}{\partial x^\mu} = F_\mu(x(p)) \tag{7.2.1}$$

I like to think of this as just a Taylor expansion. We can now move this function into one of f and then act on it with the tangent vector X_p to give us:

$$X_p = X^\mu \frac{\partial}{\partial x^\mu}|_p \tag{7.2.2}$$

This shows us that the derivatives $\partial_\mu = \frac{\partial}{\partial x^\mu}$ evaluated at any point $p \in M$ form a tangent space $T_p(M)$. It is possible to change co-ordinate systems using:

$$\frac{\partial}{\partial x^\mu}\ |_p = \frac{\partial \bar{x}^\nu}{\partial x^\mu}\ |_{\phi(p)}\ \frac{\partial}{\partial \bar{x}^\nu} \tag{7.2.3}$$

It is useful to view the tangent space as a geometrical identity inherent to the point. Co-ordinate systems just bring it down to R^n so we can use them. The tangent spaces at two points are different from each other and can't be compared directly.

7.3 Vector Fields

If we assign a tangent vector to each point on a manifold, we get a vector field. This can be defined as:

$$X = X_\mu \partial_\mu \tag{7.3.1}$$

Here, X_μ is a smooth function. We can use a vector field to differentiate over a whole function.

Now, a vector field can be viewed as a differential operator. This makes it impossible to simply multiply two vector fields X, Y to get a new, mixed vector field. One way we can get a new vector field from X, Y is to use the commutator. This is defined by:

$$[X, Y] = XY - YX = \left(X^\mu \frac{\partial Y^\mu}{\partial x^\mu} - Y^\mu \frac{\partial X^\nu}{\partial x^\mu} \right) \frac{\partial}{\partial x^\nu} \tag{7.3.2}$$

It can be verified that this commutator satisfies the Jacobi identity.

Now, it turns out that any vector field is linked to this property called a flow. We define a flow to be a one-parameter diffeomorphism, $\sigma_t : M \mapsto M$. This can be viewed as a "transformation" of sorts for the manifold. $\sigma_{t=0}$ is the identity map and σ_{-t} is the inverse of σ_t. Summing over t can give us streamlines over a manifold, these are called integral curves. If a flow is categorized by co-ordinates $x_\mu(t)$, we end up with:

$$X(x_\mu(t)) = \frac{dx_\mu(t)}{dt} \tag{7.3.3}$$

This equation basically tells us that the flow and a vector field are linked. We can integrate the vector field to get the integral curves generated by the flow, or we could differentiate the flow to get the vector field.

7.4 Push-Forward and Pull-Back

Say we have two diffeomorphic manifolds M, N with the diffeomorphism $\psi : M \mapsto N$. This can be very useful, let's look a little further into it. If we have a function defined on N, $f : N \mapsto R$, we can pull it back into the manifold M using ψ. This is done with the definition:

$$(\psi^* f)(p) = f(\psi(p)) \tag{7.4.1}$$

We can also go the other way and push an object forward. If we have a vector field Y on M, we define the push-forward using:

$$(\psi_* Y)(f) = Y(\psi^* f) \tag{7.4.2}$$

If x^μ represents a co-ordinate system on M and y^α on N, we can write the components of a pushed forward vector field as:

$$(\psi_* Y)^\alpha = Y^\mu \frac{\partial y^\alpha}{\partial x^\mu} \tag{7.4.3}$$

Now, we can choose if we want to push something forward or pull it back depending on which one is easier.
These operations are super useful. One of their uses is that they allow us to compare the tangent vectors between two points. There is nothing stopping us from just setting $N = M$, the base definition of these operations will still be satisfied. So if we know the diffeomorphism between the points on these manifolds, we can hypothetically push/pull vectors to compare them. Luckily enough, the flow we defined in the last section does exactly what we need. Thus we can combine a flow and a push-forward/pull-back to compare two tangent vectors from two distinct points!

7.5 The Lie Derivative

We know that one way of differentiating a function on a manifold is by acting on it by a vector field. However, what if we want to differentiate a vector field Y with respect to another vector field X? To achieve this, we can use the Lie derivative. We know from first principles that a derivative is defined by:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{7.5.1}$$

Using the fact that we can compare vectors of two different points using a flow and a push-forward/pull-back, we can generalize this principle to the manifold for a vector field Y:

$$\mathcal{L}_X Y = \lim_{t \to 0} \frac{((\sigma_{-t})_* Y)_p - Y_p}{t} \tag{7.5.2}$$

We are not just constrained to a vector field, we can also differentiate functions in a similar way. The lie derivative of a function just gives us: $\mathcal{L}_X f = X(f)$. This further reinforces the fact that a vector field is a fancy way of differentiating

a function defined on the manifold. The properties of the Lie derivative $\mathcal{L}_X$ are the usual properties we expect from a derivative. Using this we can simplify equation (7.5.2) further to give us:

$$\mathcal{L}_X Y = [X, Y] \tag{7.5.3}$$

It's comforting to know that the Lie derivative of a vector field with respect to another is their commutator (7.3.2). This was our initial idea for creating a mixed vector field, and the Lie derivative agrees.

7.6 One-Forms

We know that each point on the manifold has its own tangent space. To "measure" vectors defined on this tangent space, we will introduce a dual vector space (similar to the bra-ket space in Quantum Mechanics). We define a one-form such that it creates a dual vector space at each point on the manifold. If we have a one-form given defined on the basis spanned by $\{f^\mu\}$, we can write:

$$w = w_\mu f^\mu \tag{7.6.1}$$

Now since one-form's act to "measure" a vector field, we can think of the dual basis spanned by $\{f^\mu\}$ at one point as a map from the tangent space to R. One easy way to define the one-form is by the condition:

$$df(X) = X(f) \tag{7.6.2}$$

Where $f \in C^\infty(M)$ is a smooth function and X is a vector field. If we take $f = x^\mu$, where x^μ are co-ordinates on the manifold, then we find:

$$dx^\mu(\partial_\nu) = \partial_\nu(x^\mu) = \delta^\mu_\nu \tag{7.6.3}$$

This indicates that dx^μ is a suitable basis for one-forms, and we can write a general one-form as:

$$w = w_\mu dx^\mu \tag{7.6.4}$$

It is easy to verify that the one-form basis transforms like:

$$dx^\mu = \frac{\partial x^\mu}{\partial \bar{x}^\nu} d\bar{x}^\nu \tag{7.6.5}$$

It is also worth noting that acting on a one-form with a lie-derivative gives the same result as you would get for a vector field, just with the plus sign changed to a minus.

$$\mathcal{L}_X w = (X^\nu \partial_\nu w_\mu + w_\nu \partial_\mu X^\nu) dx^\mu \tag{7.6.6}$$

In summary, we can view one-forms as a way to measure the vectors defined in the tangent space. One-forms and vector fields generate a dual vector space like in quantum mechanics at each point. When we act the lie-derivative on a one-form, we get the same result as the vector field except the two terms add instead of subtract.

7.7 Tensors

We can define tensors as a map:

$$T : T_p^*(M) \times \cdots \times T_p^*(M) \times T_p(M) \times \cdots \times T_p(M) \mapsto R \tag{7.7.1}$$

The rank of a tensor (r, s) implies r, $T_p^*(M)$ spaces (which are the spaces in which one-forms reside) and s tangent spaces. A tensor field is the smooth assignment of a tensor to each point $p \in M$. We can write tensors in components, there will be n^{r+s} components for a tensor of rank (r, s). If a tensor takes a one form w and two vector fields X, Y, we can write its components as:

$$T(w, X, Y) = w_\mu X^\nu Y^\rho T(e^\mu, e_\nu, e_\rho) = w_\mu X^\nu Y^\rho T^\mu_{\nu\rho} \tag{7.7.2}$$

Another customary thing to check is how tensors transform. To check this, we will take the example of a $(2, 1)$ tensor. For the vector fields, we will consider two bases $\{e_\mu\}$ and $\{\bar{e}_\mu\}$ related to each other by:

$$\bar{e}_\nu = A^\mu_\nu e_\mu \tag{7.7.3}$$

For the one-forms, we have the respective dual bases $\{f^\mu\}$ and $\{\bar{f}^\mu\}$ which are related by:

$$\bar{f}^\rho = B^\rho_\sigma f^\sigma \tag{7.7.4}$$

By ensuring that these dual bases act to give us δ, we get the relation $B^\rho_\mu = (A^{-1})^\rho_\mu$. Thus, the $(2, 1)$ tensor transforms as:

$$\bar{T}^\mu_{\rho\nu} = B^\mu_\sigma A^\tau_\rho A^\lambda_\nu T^\sigma_{\tau\lambda} \tag{7.7.5}$$

This relation is pretty important. If an object transforms like equation (7.7.5), it can be called a tensor. This is why people define tensors as objects that transform like a tensor.

We can now look at operations with tensors. The first being the tensor product. Given a tensor S of rank (p, q) and a tensor T of rank (r, s) we can construct a tensor $S \otimes T$ of rank $(p + r, q + s)$ defined by:

$$\begin{aligned} &S \otimes T(w_1, ..., w_p, \eta_1, ..., \eta_r, X_1, ...X_q, Y_1, ...Y_s) \\ &= S(w_1, ...w_p, X_1, ...X_q)\, T(\eta_1, ..., \eta_r, Y_1, ..., Y_s) \end{aligned} \tag{7.7.6}$$

The tensor product in terms of components is given by:

$$(S \otimes T)^{\mu_1 \ldots \mu_p \nu_1 \ldots \nu_r}_{\rho_1 \ldots \rho_q \sigma_1 \ldots \sigma_s} = S^{\mu_1 \ldots \mu_p}_{\rho_1 \ldots \rho_q} T^{\nu_1 \ldots \nu_r}_{\sigma_1 \ldots \sigma_s} \tag{7.7.7}$$

In addition to being multiplied, we can also make tensors symmetric or anti-symmetric. We will take a rank $(0, 2)$ tensor as an example:

$$S_{\mu\nu} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu}) \ ; \ A = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}) \tag{7.7.8}$$

8 Appendix B: Riemannian Geometry

Riemannian geometry is just more interesting differential geometry. This discusses stuff that happens to/on a manifold when we introduce a metric. Again, this discussion is not meant to be rigorous. The only purpose of this discussion is to provide a quick summary.

8.1 The Metric

We start this appendix by discussing the metric. The metric is how we define the inner product on a manifold. We define the metric as a $(0, 2)$ tensor field that is:

1. Symmetric, $g(X, Y) = g(Y, X)$
2. Non-Degenerate, if for any $p \in M$, $g(X.Y) \mid_p = 0$ for all $Y \in T_p(M)$ then $X_p = 0$

We can create this tensor by taking the direct product of two $(1, 0)$ elements which are one-forms. We need to have two one-forms because by definition we want to map vector fields to R. Thus we can write:

$$g = g_{\mu\nu}\ dx^\mu \otimes dx^\nu \tag{8.1.1}$$

This is also generally written as:

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu \tag{8.1.2}$$

We can find the components of the metric by evaluating it in the tangent basis.

$$g_{\mu\nu} = g\left(\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}\right) \tag{8.1.3}$$

We generally try to pick the basis where the metric is diagonal.
A Riemannian manifold is one where all entries of the manifold are positive. We can use the metric here to measure the distance along a curve using:

$$l = \int_a^b dt \sqrt{g(X, X) \mid_{\sigma(t)}} \tag{8.1.4}$$

Where, $\sigma(t)$ parameterizes the curve. This makes physical sense as the metric is supposed to represent the inner product of two vectors. If we take the inner product of a vector with itself we get its magnitude, and since X is the tangent vector, integrating that will just give us the distance. Since the metric is the inner product, we can also use it to measure the angle between two vectors.
Now, the whole purpose of this math is to apply it to relativity. We know that in Minkowski space, we have a negative diagonal entry to the metric. Manifolds with a metric like this are called Lorentzian manifolds. Curve distance is measured the same way for Lorentzian manifolds as for the Riemannian manifolds.

Now, for each point $p \in M$ it is possible to locally pick an orthonormal basis such that the metric resembles that for Minkowski space. This is done using normal coordinates. This allows us to translate certain concepts from special relativity (such as light/time/space-like particles) to general relativity locally. Locally is an important point to remember here.

8.1.1 The Metric as a Natural Isomorphism

Say we have the metric component $g_{\mu\nu}$. If we act this on the vector field X we get $g_{\mu\nu}X^\mu$. This term acts as a map: $T_p(M) \mapsto R$. This is however, the same thing a one-form does. This tells us that the metric can act as an isomorphism between vectors and covectors $g : T_p(M) \mapsto T_p^*(M)$. We can now introduce some new notation, $X = X_\mu dx^\mu$ represents the one-form while $X = X^\nu \partial_\nu$ represents the vector field. So we get:

$$X_\mu = g_{\mu\nu}X^\nu \tag{8.1.5}$$

The inverse metric does the inverse (shocker).

$$X^\mu = g^{\mu\nu}X_\nu \tag{8.1.6}$$

8.1.2 The Metric as a Natural Volume Form

We can use the metric to define a volume form:

$$v = \sqrt{detg_{\mu\nu}}\ dx^1 \wedge \cdots \wedge dx^n \tag{8.1.7}$$

Its standard notation to contract the determinant as $\sqrt{detg_{\mu\nu}} = \sqrt{g}$. The determinant will be negative for a Lorentzian metric.
This volume-form can be verified to be independent of coordinate systems. To explore the physical meaning of this volume form, we first express it in the form:

$$v = \frac{1}{n!}v_{\mu_1\cdots\mu_n}dx^{\mu_1} \wedge \cdots dx^{\mu_n} \tag{8.1.8}$$

Here,

$$v_{\mu_1\cdots\mu_n} = \sqrt{g}\epsilon_{\mu_1\cdots\mu_n} \tag{8.1.9}$$

We have defined the ϵ as the anti-symmetric tensor density with $\epsilon_{1\cdots n} = 1$. This tensor density should be viewed as a volume-form on the manifold. We can say that the metric allows us to weigh certain regions of the manifolds differently for integration.
Now, we can integrate any function over a manifold using:

$$\int_M fv = \int_M d^n x\sqrt{g}\ f \tag{8.1.10}$$

8.2 Torsion and Riemann Tensor

We can define a few tensors using the covariant derivative. The first of these is a rank $(1, 2)$ tensor known as torsion. It is defined by:

$$T(w; X, Y) = w(\nabla_X Y - \nabla_Y X - [X, Y]) \tag{8.2.1}$$

The second is the Riemann tensor. This is a rank $(1,3)$ tensor defined by:

$$R(w; X, Y, Z) = w(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z) \tag{8.2.2}$$

This tensor is also known as the Riemann curvature tensor, it encodes the curvature of spacetime and the path-dependence of parallel transport. If we remove the one-form from the definition of torsion, we get map that acts like $\mathcal{X}(M) \times \mathcal{X}(M) \mapsto \mathcal{X}(M)$ defined by:

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y] \tag{8.2.3}$$

Under a similar action, the Riemann tensor becomes a differential operator:

$$R(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]} \tag{8.2.4}$$

We can also look at these tensors in coordinates. The torsion tensor is given by:

$$T^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\mu\nu} - \Gamma^{\rho}_{\mu\nu} \tag{8.2.5}$$

This tensor is anti-symmetric in its bottom indices. Torsion free also implies $\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu}$. The Riemann tensor is given by:

$$R^{\sigma}_{\rho\mu\nu} = \partial_\mu \Gamma^{\sigma}_{\nu\rho} - \partial_\nu \Gamma^{\sigma}_{\mu\rho} + \Gamma^{\lambda}_{\nu\rho} \Gamma^{\sigma}_{\mu\lambda} - \Gamma^{\lambda}_{\mu\rho} \Gamma^{\sigma}_{\nu\lambda} \tag{8.2.6}$$

This tensor is anti-symmetric in its last two indices.

$$R^{\sigma}_{\rho\mu\nu} = -R^{\sigma}_{\rho\nu\mu} \tag{8.2.7}$$

8.3 The Levi-Civita Connection

If we have both a metric and a connection ∇. We can apply what is known as the fundamental theorem of Riemannian geometry. This states that there exists a unique, torsion free, connection that is compatible with a metric. Mathematically:

$$\nabla_X g = 0 \tag{8.3.1}$$

For all vector fields X. We can try to compute its components in a coordinate basis $\{e_\mu\} = \{\partial_\mu\}$. This gives us:

$$g(\nabla_\nu e_\mu, e_\rho) = \Gamma^{\lambda}_{\nu\mu} g_{\lambda\rho} = \frac{1}{2}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}) \tag{8.3.2}$$

If we multiply this equation through my the inverse metric, we can identify the components of the Levi-Civita connection as being the Christoffel symbols. In General Relativity, we mostly consider all connections to be Levi-Civita connections.

8.4 Normal Coordinates

These coordinates are pretty useful. Essentially, on a Riemannian manifold, in the neighbourhood of a point $p \in M$ we can always find coordinates such that:

$$g_{\mu\nu}(p) = \delta_{\mu\nu} \ ; \ g_{\mu\nu,\rho}(p) = 0 \tag{8.4.1}$$

These coordinates hold for Lorentzian manifolds as well and are known as normal coordinates. We can interpret these coordinates as "rotating" our reference frame to make spacetime look flat at point p.
These coordinates can be computed either explicitly or by using a concept known as an exponential map.

8.5 Ricci and Einstein Tensors

We can actually contract two-indices on a Riemann tensor to give us the Ricci tensor which is of rank $(0, 2)$. Mathematically this is given by:

$$R_{\mu\nu} = R^{\rho}_{\rho\mu\nu} \tag{8.5.1}$$

This Ricci tensor can also be used to construct a function over a manifold known as the Ricci scalar.

$$R = g^{\mu\nu} R_{\mu\nu} \tag{8.5.2}$$

The Ricci tensor also inherits its symmetry from the Riemann tensor.

$$R_{\mu\nu} = R_{\nu\mu} \tag{8.5.3}$$

An interesting fact about the Riemann and Ricci tensors is that they encode the change in a volume element as we move through a manifold. This can be traced to the fact that these tensors encode path dependence of a vector. So by combining path dependence in multiple directions, we can judge how a volume element shrinks or expands. This gives a new meaning to Ricci flat spacetime. This type of spacetime can be interpreted as one in which the volume element stays the same size. We can also define the Einstein tensor.

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \tag{8.5.4}$$

Importantly, the Einstein tensor is covariantly conserved.

$$\nabla^{\mu} G_{\mu\nu} = 0 \tag{8.5.5}$$

9 Interesting Topics

This is a list of interesting patterns and questions that emerged in the course of writing this project document.

1. Why does the differentiation of a one-form and vector field differ only by one sign for both the lie derivative and covariant derivative?

2. Why do we need actions to be second-order in the derivative?